Toll of Trolls: Are Politicians Voluntarily Exiting Office Because of Online Hostility?

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Abstract

Are politicians voluntarily exiting office because of online abuse? I answer this question using nearly a decade of observational panel data on over a thousand US politicians. By connecting how much hostility politicians receive on a social media platform with their subsequent campaign decisions, I overturn the dominant expectation that online hostility is disproportionately driving women to exit politics. I find that exposure to hostility is not associated with a woman's campaign trajectory but substantially increases a man's likelihood of voluntary exit. Women who faced more gendered barriers in entering office are the least likely to voluntarily exit in relation to abuse, pointing toward a gender-based selection effect where only the most thick-skinned women pursue political careers. If online hostility is disproportionately decreasing women's descriptive representation, it is likely doing so at the decision to enter rather than exit office.

Keywords: online hostility, state legislatures, ambition, violence against women in politics, social media, representation

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There is broad speculation that abuse on social media platforms may lead women politicians to voluntarily exit elective office at higher rates than men (Carson et al. 2024; Erikson et al. 2023; Harmer and Southern 2023; Herrick and Franklin 2019; Herrick and Thomas 2022; Herrick et al. 2026; Håkansson 2024; Håkansson and Lajevardi 2024; Jankowicz 2017; Jarman 2026; Kosiara-Pedersen 2023; Krook 2017, 2020; Sobieraj 2018; Thomas et al. 2019; Yan and Bernhard 2024). These conjectures build on two important pieces of evidence. The first is that women in politics receive incommensurate gender-based hostility online, creating a unique and plausibly more abusive digital environment than men encounter (Erikson et al. 2023; Gorrell et al. 2020; Jarman 2026; Southern and Harmer 2019). The second is that women politicians are more likely than men to report behavioral changes following exposure to abuse (Erikson et al. 2023; Herrick and Franklin 2019; Herrick et al. 2026; Håkansson and Lajevardi 2024; Ramachandran et al. 2024). Despite the widely echoed concern, it remains unknown whether politicians—men or women—are in fact curtailing their political careers because of online abuse.

I employ gender-based selection theory to propose a competing expectation to the general prediction. Gender-based selection theory suggests that sex discrimination in the electoral process—whether perceived or actualized—deters all but the most qualified and ambitious women from emerging as candidates (Anzia and Berry 2011). As a result, women politicians outperform their men colleagues despite the gendered hurdles they face (Anzia and Berry 2011; Bauer 2020b; Bauer and Cargile 2023; Butler et al. 2022; Eatough and Preece 2025; Fulton 2012; Lazarus and Steigerwalt 2018; McGrath et al. 2025; Schmitt and Brant 2019; Thomsen and Sanders 2020). The same logic implies that women officeholders should be as resilient to online hostility as they are to other gendered barriers.

This study identifies the effect of online hostility on voluntary political exit using nearly a decade of observational panel data. The dataset includes the campaign decisions of over 1,200 lower-house American state representatives (370 women, 913 men) across four election cycles (2015–2023). State legislators are the ideal case study as they have fewer resources to screen abuse and are well-placed to consider movement across offices (Butcher 2021; Butler et al. 2023; Holman 2017; McCrain and O'Connell 2023). I additionally collect all concurrent public tweets referencing these politicians. I identify hostile and gendered content in the public tweets to measure how much—and what type of—online hostility each politician encountered in the six months leading up to the deadline to file to campaign. By linking these datasets, I estimate whether exposure to online hostility influenced subsequent decisions to campaign for reelection, seek a different office, or voluntarily exit.

Contrary to the general expectation that online hostility is disproportionately pushing women politicians to exit office, I find that the effects of online hostility are concentrated

among men. Increased exposure to hostility substantially increases a man's likelihood of voluntary exit, but is not associated with a woman's campaign choices. This pattern holds even when accounting for gender-based hostility and the severity of abuse. Women politicians who faced more gendered barriers in entering office are the least likely to voluntarily exit in relation to online hostility, indicating support for gender-based selection.

More than 80% of politicians report experiences of online abuse, and there is substantial evidence that online violence against politicians is increasing (Lai et al. 2022; Morton et al. 2023; Ramachandran et al. 2024; Simi et al. 2024). As the first paper to identify the consequences of online hostility for subsequent campaign decisions, this paper addresses an important gap in the literature. The results yield two important findings. First, abuse toward politicians on social media platforms is a severe form of political violence that influences who stays in office and who exits. Second, women who successfully attain office are not uniquely brittle to this abuse. The findings imply that if online hostility disproportionately reduces women's descriptive representation, it likely does so by discouraging entry rather than driving exit.

Why do politicians voluntarily exit?

Gender plays a major role in shaping who decides to enter politics. Demand-side research identifies gendered hurdles in recruitment (Crowder-Meyer 2013; Karpowitz et al. 2017; Ono and Endo 2024) and bias against women candidates (Bateson 2020; Hassell and Visalvanich 2024; Mo 2015; Saha and Weeks 2022). On the supply side, women entrants are inhibited by gendered socialization (Bos et al. 2022; Kanthak and Woon 2015; Preece and Stoddard 2015), self-perceived qualifications (Crockett 2019; Fox and Lawless 2024; Pate and Fox 2018), and structural barriers (Bernhard et al. 2021).

Gender may play a decisive role in the decision to enter politics, however, once in office, it does not appear to influence the decision to exit politics. Gender does not appear to mediate static or progressive ambition, respectively defined as the desire for a long career in a single office and the aspiration for a more powerful position (Schlesinger 1966). Once in office, men and women are similarly likely to campaign for reelection (Butcher and Haynes 2024; Dittmar 2024; Dittmar and Shah 2025; Thomsen 2017) and a higher office (Butcher 2021; Fulton et al. 2006; Maestas et al. 2006).[1] Within state legislatures, gender does not have a statistically significant effect on retirement, resignation, electoral loss, or leaving to pursue higher office (Butcher 2021). Furthermore, among narrow losers of elections, women

[1] See Brown et al. (2019) for countervailing evidence that a lower proportion of women than men advance to national office.

are as likely as men to campaign again (Bernhard and de Benedictis-Kessner 2021).

Instead, scholars have explained political exit using legislative design and party strength. Individuals elected to more professionalized chambers with clearer career ladders are less likely to exit politics (Butcher 2021; Butcher and Haynes 2024; Kanthak 2011). Strong parties can also encourage longer tenures by creating internal promotion pathways for ideologically aligned members (Butcher 2023). Ideological moderates, on the other hand, perceive fewer opportunities for advancement (Kanthak 2011) and are less likely to seek higher office or reelection (Hall 2019; Thomsen 2017).[2]

Increasingly, legislators are choosing to not exit, instead aspiring to remain in elected office until retirement (Mattozzi and Merlo 2008; Roberts 2019). Historically, many U.S. politicians left politics before retirement to work in the private sector (Diermeier et al. 2005). In the past two decades, however, the average turnover rate has decreased in all but one of the U.S. state legislative chambers (Butcher 2022). This change is partially explained by the implementation of term limits, which have unintentionally lengthened politicians' tenure. In states with term limits, both men and women are more likely to remain in office until they are forced out rather than retire voluntarily (Butcher 2021).[3]

Online hostility may encourage exit

As online hostility toward politicians becomes ever more prevalent—more than 80% of politicians report experiences of online abuse—scholars have sought to identify whether it is reshaping the utility calculus of holding office (Lai et al. 2022; Morton et al. 2023; Ramachandran et al. 2024; Simi et al. 2024). Campaign decisions, including exit, are often framed as a calculation of costs and benefits (Black 1972). Online hostility may encourage politicians to exit by increasing the costs of holding office (Carson et al. 2024; Erikson et al. 2023; Harmer and Southern 2023; Wagner 2022). Recipients of online hostility report greater symptoms of psychological distress, including higher anxiety, greater concern for their loved ones, and worsened sleep (Herrick and Franklin 2019; IPU 2016, 2018). Affected politicians also report a range of practical costs, such as the financial strain of paying for increased security (Bridging Divides Initiative 2025; IPU 2016; Morton et al. 2023; Munger 2017; Ramachandran et al. 2024). The choice to speak out about online abuse comes with its own set of consequences. Politicians who respond to online harassment are accused of dramatizing

[2] Thomsen (2025) argues that early fundraising success is used as a meaningful signal of candidate viability by both the candidates themselves as well as those attentive to the race. However, because Thomsen (2025) focuses on the drop-out decisions of active candidates, it is beyond the scope of this paper.

[3] Because women hold fewer seats overall, term limits have a greater relative effect on reducing women's exit rates (Butcher and Haynes 2024).

the severity of the abuse and viewed as weaker candidates (Håkansson and Grahn 2025; Schechet 2017).

Online hostility may be depleting the benefits of holding office at the same time that it escalates the costs. Exposure to abuse changes how politicians perform their job, particularly in how they engage with their constituents. Many affected politicians decrease their political presence by holding fewer in-person events and reducing their overall contact with citizens (Bridging Divides Initiative 2025; Ramachandran et al. 2024). In seeking to mitigate the hostility, some legislators reevaluate which topics they feel safe addressing (Celuch et al. 2024; Håkansson 2024; Ramachandran et al. 2024). By limiting their legislative substance and social connection to constituents, online hostility may decrease the benefits of holding office for some politicians.

Are women disproportionately affected?

The costs of online hostility are not evenly distributed between men and women politicians, leading to widespread concerns that online hostility may be asymmetrically pushing women from office. Compared to men, women politicians receive gendered forms of hostility online (Bardall et al. 2020). Although men and women experience similar levels of general hostility online, abuse toward women is twice as likely to be gender-based in content (Jarman 2026). In particular, women politicians self-report significantly higher rates of sexualized harassment, gender-based hate speech, and sexist abuse on social media platforms than men (Bjarnegård 2018; Erikson et al. 2023; Esposito and Breeze 2022; Herrick et al. 2022; Holm et al. 2024; Kosiara-Pedersen 2023). These self-reports are supported by computational text analyses which account for such gendered content (Gorrell et al. 2020; Jarman 2026; Ward and McLoughlin 2020).

In addition to taking gendered forms, online abuse toward politicians elicits gendered impacts. Women are more likely than men to self-report reducing their public presence and censoring what they say following online attacks on themselves or their peers (Herrick et al. 2026; Holm et al. 2024; Håkansson and Lajevardi 2024; Jankowicz 2017; Pedersen et al. 2024; Ramachandran et al. 2024; Sobieraj 2020). Among state legislators, targeted women report greater reluctance to use social media, make traditional media appearances, give interviews, and hold public events (Ramachandran et al. 2024). Most compellingly, larger shares of women than men report considering exiting politics after experiencing online attacks (Erikson et al. 2023; Herrick and Franklin 2019; Håkansson and Lajevardi 2024; Ramachandran et al. 2024).

The finding that women receive disproportionately gendered online abuse, paired with

their higher propensity for reporting effects of online abuse, has cultivated a sense that online hostility is asymmetrically pushing women politicians to exit office. This conjecture has been echoed in over a dozen publications across research on online hostility, gendered violence, and violence against women in politics (Carson et al. 2024; CCDH 2024; Harmer and Southern 2023; Herrick and Franklin 2019; Herrick and Thomas 2022; Herrick et al. 2026; Håkansson 2024; Håkansson and Lajevardi 2024; Jankowicz 2017, 2022; Kosiara-Pedersen 2023; Krook 2017, 2020; Ramachandran et al. 2024; Sobieraj 2018, 2020; Thomas et al. 2019; Wagner 2022). An aim of this paper is to empirically test this hypothesized relationship between abuse and exit.

> **H1:** Exposure to hostility will disproportionately increase women politicians' likelihood of voluntary exit relative to men.

Selection effects may counter online hostility

Without challenging the patterns of online hostility's gendered forms and impact, there are still reasons to doubt the expectation that women politicians would be disproportionately affected. Chief among these arguments is that women politicians have repeatedly demonstrated the ability to excel despite gendered barriers. Indeed, gendered online hostility is one of many additional impediments that women politicians experience due to their gender. Compared to men, women politicians report greater difficulty fundraising (Barber et al. 2016; Jenkins 2007; Thomsen and Swers 2017), receive less valuable committee seats than men (McGrath et al. 2025), and are more likely to be targeted by challengers (Lazarus and Steigerwalt 2018). Across the board, women perceive higher costs to holding office than men do (Fulton et al. 2006; Shames 2017).

Yet, women politicians overcome these gendered costs to continue serving in elective office. Women raise as much money as men (Hayes and Lawless 2016), are more effective at passing legislation (Eatough and Preece 2025), and demonstrate comparable rates of reelection as men (Butcher 2021, 2022, 2023; Butcher and Haynes 2024; Thomsen and Swers 2017). In addition, women are more responsive to constituents (Bauer and Cargile 2023; Butler et al. 2022; Dickson 2025; Thomsen and King 2020) and cultivate more comprehensive legislative agendas (Butler et al. 2022; Lazarus and Steigerwalt 2018; Schmitt and Brant 2019).

The theory of gender-based selection explains the duality of women's simultaneous greater challenges and performances. The theory proposes two pathways for gender-based selection. The expectation of sex discrimination deters all but the most qualified and ambitious women from emerging, while actualized voter bias further narrows which women can attain office

(Anzia and Berry 2011). As a result, the pattern of "when women run, women win" masks a reality where women must often exceed their male counterparts in demonstrable quality to be perceived as equally competitive (Bauer 2020a; Fulton 2012; Lazarus and Steigerwalt 2018; Pearson and McGhee 2013).

There is considerable evidence that women politicians include online hostility as a cost of holding office prior to candidacy. More so than men, women citizens and political aspirants anticipate that campaigning for office would expose them to abuse and discrimination (Fox and Lawless 2024; Kjøller and Pedersen 2025; Shames 2017). Presumably, the women who did emerge also foresaw hostility. This assumption is supported by the strategies that candidate training groups advocate women use to prepare for online abuse: digital security tools to prevent hacks and doxxing, hiring staff to manage social media accounts, and building a community of similarly-targeted women to help report offensive comments and advise on legal action (Jankowicz 2022; Schriock and Reynolds 2021; She Should Run 2022; Sobieraj 2020; Tenove et al. 2023). Women who entered political office prior to widespread social media use still expected sex discrimination, even if they were unfamiliar with the format which has since become dominant (Anzia and Berry 2011). If women in the candidate pool are aware of the hostility women politicians experience, those who choose to pursue office will be exceptionally resilient to this abuse, perhaps even more so than men.

Gender-based selection theory further implies that women politicians will demonstrate adaptive strategies for tolerating hostility - an implication which is supported in the literature. Some women normalize the abuse while others "out" their attackers or use insulting language as a means of self-defense (Astor 2018; Boukemia et al. 2025; Phillips 2017; Sobieraj 2020). Many women extensively cite official sources in their online communication to prevent attempts at discrediting them (Jankowicz 2022; Schriock and Reynolds 2021; Sobieraj 2020). Rather than be evidence that online hostility is pushing women out of office, the behavioral impacts reported earlier—reducing public exposure, limiting engagement, and avoiding volatile issues—may help women stay in office longer by staving off compounded hostility (Celuch et al. 2024; Herrick et al. 2026; Ramachandran et al. 2024).

Although the gender-based selection literature has not directly engaged with the question of how women politicians will respond to online hostility, it intimates a competing expectation of the online hostility and violence against women in politics literatures. As a result, I posit the following:

> **H2:** Exposure to hostility will *not* disproportionately increase women politicians' likelihood of voluntary exit relative to men.

The extension of gender-based selection theory to online VAWiP has observable impli-

cations. Namely, women who had to overcome additional challenges to enter political office should be more resilient to abuse than their counterparts. This leads to the final hypothesis:

> **H3:** The more gendered barriers women faced in entering office, the less likely they will be to voluntarily exit politics in relation to online hostility.

I test these hypotheses using observational data. I collect both the career trajectories and online environments of politicians spanning nearly a decade. By pairing these data, I am able to identify the comparative influence of online hostility on men and women politicians' exit and retention. In Part 1 of the analyses, I consider the comparable evidence for the mutually exclusive **H1** and **H2** about the effect of online hostility on voluntary exit. In Part 2, I evaluate the role of gender-based selection in the association between hostility and campaign choice (**H3**).

Identifying movement within or out of office

In determining the influence of exposure on ambition, I choose to use US state legislators as my case study for three reasons. First, state legislators have fewer resources to screen abuse, which increases the likelihood of violence making direct contact with the targeted politicians (Butler et al. 2023; Wood 2021).[4] Second, most national politicians begin their career in state legislatures (Holman 2017; Manning 2024; McCrain and O'Connell 2023). State legislators have already demonstrated political ambition by entering office, yet many are still early in their careers. Finally, prior research illustrates that gender does not have a statistically significant effect on retirement, resignation, electoral loss, or leaving to pursue office among state legislatures (Butcher 2021). Thus, holding exposure to hostility constant, I do not expect gender to independently influence a politician's decision to exit.

I collected the career paths of 1,247 politicians who were all lower-house state representatives in 2017.[5] I gathered their campaign decisions and outcomes from 2017 through 2023 using data from the Center for American Women and Politics (CAWP) and Ballotpedia. At the end of each politician's term, I documented whether they chose to campaign for reelection, a different office, or not at all. If they chose to campaign, I tracked which campaigns the politician entered, the state of the primary and general race, and the election

[4] National politicians often hire staff whose job is to curate a daily "social media scrap" of relevant activity for their employer. As a result, well-resourced politicians are able to avoid direct engagement with abusive content online.

[5] Note: The original intention was to scrape Twitter mentions for the 3,399 state representatives identified by Butler et al. (2023) who were active on Twitter in 2017. As this process occurred, Twitter discontinued its academic API, cutting off the data collection at 1,247 state representatives.

result. I also accounted for whether the politician received an appointed position or was term-limited out of their office. If a politician reached the end of their term and did not campaign nor assume an appointed position, she is considered to exit. I continued to collect campaign decisions for politicians who exited to account for political rebirths. The resulting dataset has 4,864 observations of politician-election year pairs. The demographic data of each politician—including race, party, gender, and district—were collected by Butler et al. (2023).

Candidates must file paperwork by a specific deadline to appear on the ballot in their state. Deadlines vary by state and year. I collected filing deadlines from Ballotpedia and National Conference of State Legislatures. Politicians may determine which races to enter long before the filing deadline, however, the filing deadline represents a hard cut-off for the decision to campaign or exit. By adding the filing deadlines to the career path dataset, I can approximate a decision timeline for each politician – election year pair. I use the filing deadlines to form summary measures of a politician's online environment in the six months leading up to the decision cut-off.[6]

As all the individuals held a state representative office in 2017, I filtered the dataset so 2018 is the first included election cycle. For legislatures with a two-year term, the data thus encompasses three election cycles. For individuals who exit and do not campaign again, only the first cycle they sat out is labeled as an exit. Subsequent inactive years are not considered in the model. Similarly, politician - election year pairs where the individual is term-limited out of office are not included in the dataset.

Table 1 shows the breakdown of politician campaign choices by gender across all politician-election year pairs. While the most common choice is to campaign for reelection to the same office, more than 20% of the decisions are to exit. Movements toward a different office are comparatively infrequent. Gender alone does not significantly influence the likelihood of campaigning for reelection, a different office, or not at all.[7] This matches results previously shown in Butcher (2021).

Table 1: Politicians prefer to campaign for reelection

	Reelection	**Exit**	**Different Office**
Women	756 (68.2%)	207 (20.8%)	108 (10.91%)
Men	1581 (67.9%)	562 (24.2%)	184 (7.9%)

[6] In the appendix, I compare the six month timeline to alternate windows of two-and-a-half years and the accumulative hostility received up to that filing deadline (see Table 4).

[7] See Table 3.

Identifying gendered hostility online

I scraped all the mentions toward the same 1,247 state representatives that were available on Twitter in February 2023 and originated after June 2015. Mentions are public tweets which tag the legislator by including their handle. The Twitter handles for the representatives were collected by Butler et al. (2023). As the scraping process occurred, Twitter discontinued its academic API. The list of politicians to scrape were ordered alphabetically by first name. As a result, the dataset is limited to politicians whose first names begin with the letter 'a' through 'i'. Balance tests between the scraped and un-scraped politicians do not reveal proportional differences across gender, party, or race.

At the time the handles were identified, women and Democratic representatives were more likely to have a Twitter account than men or Republican representatives.[8] The resulting corpus has approximately 14 million tweets directed at 370 women and 913 men politicians over four election cycles.[9] Approximately half of the politicians are Republican, although women are nearly twice as likely to be members of the Democratic party. Most of the politicians (79%) are white. Table 23 describes the composition of the sample.

I then trained a machine learning classifier to label the text content of each tweet. Training labels are produced using a combination of human coding and zero-shot LLM annotations.[10] Zero-shot annotations are a "fast, efficient, and cost-effective" way to improve the performance of supervised classification algorithms (Møller et al. 2024; Pangakis and Wolken 2024). LLM annotations are validated against human coders and indicate strong performance, achieving F1 scores between 0.7 and 0.83. A full comparison of the zero-shot and human-coded labels is available in the Supplementary Materials.

The expert coders and LLM received identical labeling instructions, which can be viewed fully in the Supplementary Materials. Each mention was classified separately for hostile, gendered, and supportive content. Briefly, a mention is "hostile" if it contains impolite, rude, derogatory comments, vulgarities, threats, hate speech, dismissive tones, sexual harassment, racism, ad hominem attacks, or calls for a legislator's resignation. A mention is "gendered" if it references a legislator's physical appearance, relationship status, parental role, competence based on gender, utilizes gendered pejoratives, comments on stereotyped gender traits, or directly mentions the legislator's gender. A mention is "supportive" if it explicitly supports

[8] Liberals were overrepresented among Twitter users during this time frame (Wojcik and Hughes 2019). According to a national survey of adults in 2018, 36% of Twitter users were Democrat, 21% were Republican, and 29% were independent. Among the general population, 30% of U.S. adults were Democrat, 26% were Republican, and 27% were independent.

[9] Most state legislatures have two-year terms.

[10] I use `gpt-4-turbo`.

the legislator, encourages others to support the legislator, or praises the legislator.

Table 2 shows a randomly chosen tweet from each combination of labels. These confirm that the LLM accurately detects hostile, gendered, and supportive content.[11] I applied design-based supervised learning to correct errors in the BERT predictions to match the hand-coded labels (Egami et al. 2024).

Table 2: Labels show high face validity

Example Tweet	**Hostile**	**Gendered**	**Supportive**
"@legislator Really have to be the stupidest person on Twitter don't you?"	1	0	0
"@legislator You are in way over your head little girl!!"	1	1	0
"congressman @legislator gives his take on HB112"	0	1	0
"Women like @legislator are authentically themselves in representing the American people"	0	1	1
"@legislator Thank you for all that you're doing!!"	0	0	1

In addition, I validate the BERT-classified labels using the open-source model Detoxify. Detoxify is a machine learning model that calculates the probability that text includes the following: toxicity, severe toxicity, obscene, threat, insult, identity attack, sexually explicit language.[12] Each mention in the dataset is scored on the probability of containing each of these dimensions. As expected, I find that hostile-labeled mentions score much higher in each abuse category than supportive or neutral mentions (see Figure 6).

[11] The F1 scores between the BERT predictions and the hand-coded tweets are 0.88 for gendered, 0.78 for hostile, and 0.83 for supportive.

[12] The training data for this model was annotated in 2018 on the Figure Eight crowd rating platform, with each tweet being annotated by up to ten workers. The 'unbiased' model corrects for the tendency of text-classification models to assume all mentions of identities are intended negatively. Toxicity is defined as "a rude, disrespectful, or unreasonable comment that is somewhat likely to make you leave a discussion or give up on sharing your perspective". Severe toxicity follows the same schema, with the difference of *very* likely as opposed to *somewhat*. The category of identity attack specifically considers gender (male/female), sexuality, religion, race, and psychiatric or mental illness. More information is available on Unitary AI's GitHub page.

Part 1: Are women disproportionately exiting because of hostility?

This section tests two theoretically distinct and mutually exclusive hypotheses. **H1** posits that exposure to hostility will disproportionately increase women politicians' likelihood of voluntary exit relative to men. In contrast, **H2** predicts that exposure to hostility will *not* disproportionately increase women politicians' likelihood of voluntary exit relative to men.

My chief explanatory variable is the proportion of total hostility—both general and gendered hostility—a politician received in the six months prior to the filing deadline to campaign. I approximate politician quality using five indicators: legislator effectiveness,[13] the year they were first elected to office,[14] lagged fundraising,[15] and visibility.[16] In addition, I account for the legislator's gender,[17] partisan affiliation,[18] whether the legislator is a person of color,[19] and if they hold an elective position at the time of the filing deadline. Lastly, I incorporate Ladam et al. (2018)'s measure of women political culture in each state in 2012.

The dependent variable—campaign choice—is categorical, with three possible outcomes: voluntarily exiting office, campaigning for reelection, or campaigning for a different office than the one a legislator currently holds.[20] Voluntary exit serves as the reference category. The multinomial logit model therefore estimates whether greater exposure to hostility shifts legislators toward exit rather than either remaining in their current office or campaigning for a different one. Relative risk ratios above one in Table 3 indicate an increase in the likelihood of the respective outcome relative to exit, while ratios below one indicate a greater likelihood of exit. Standard errors are based on 10,000 bootstrap replications clustered at the individual

[13] The State Legislative Effectiveness Score captures the number of bills a legislator sponsored, the extent to which those bills advanced through the legislative process, and their substantive importance (Bucchianeri et al. 2025). I use the 2017 SLES score because all the legislators in the dataset held office during this year. Thomsen et al. (2026) finds that SLES scores correspond to progressive ambition among less professional legislatures but not in more professional legislatures.

[14] I collected the first-year an individual was elected to any office using Ballotpedia, Wikipedia, personal websites, and newspaper articles.

[15] I incorporate fundraising information from Follow The Money https://www.followthemoney.org/our-data/about-our-data. This lagged variable provides the total sum raised by a candidate in their previous campaign. Where possible, I supplement fundraising data with DIME (Bonica 2024).

[16] I use the number of mentions a politician received on Twitter as a measure of their name recognition and visibility to the public (Cha et al. 2010; Gorrell et al. 2020; Jarman 2026; Rheault et al. 2019; Theocharis et al. 2020; Ward and McLoughlin 2020).

[17] Gender identity was collected by Butler et al. (2023) and verified where possible with CAWP (2024).

[18] Party affiliation was collected for each election cycle using CAWP (2024) and Ballotpedia.

[19] Race and ethnicity were collected by Butler et al. (2023).

[20] The appendixes include two alternate specifications for the dependent variable: a binomial logit for campaigning, and a multinomial logit that distinguishes between campaigning for a higher or lower office than the politician currently holds (see Table 8 and Table 10). Both models yield similar findings to those shown below.

level.

I test the robustness of the model results using minimum detection effects, equivalence testing, and alternating the measure of hostility based on content, severity, and time under consideration. Finally, I assess whether political ambition influences the reception of online hostility. Across all the results, I do not find evidence of the disproportionate effect of hostility on women's' campaign choices that **H1** predicted. I do, however, find evidence consistent with **H2**.

Women are not asymmetrically exiting

Contrary to the widespread expectation that online hostility is disproportionately pushing women politicians to exit office, I find that the effects of online hostility on campaign choice are concentrated among men. Among men, greater total hostility is associated with substantially higher odds of voluntarily exiting rather than campaigning for reelection ($RRR = 0.27, p < 0.05$) or another office ($RRR = 0.14, p < 0.05$). In contrast, total hostility is not significantly correlated with women's decision to campaign for reelection, run for a different office, or voluntarily exit. Although greater exposure to hostility is associated with a higher likelihood of voluntary exit among men but not women, the interaction between Total Hostility and Woman is not statistically significant in Table 3. There is no evidence of the disproportionate exit predicted by **H1**.

Using ten thousand cluster-bootstrap samples, I calculate each politician's predicted probability of each campaign outcome at their observed level of total hostility and after increasing their hostility by twenty-five percentage points. This increment represents the difference between the 25th and 75th percentiles of total hostility among legislators exposed to hostility. I take the difference between those two predicted probabilities and then average those differences separately for men and women. The resulting Figure 1 illustrates the average change in the predicted probability of each outcome when total hostility meaningfully increases.

Among men, an interquartile increase in total hostility is associated with an approximately five percentage point increase in the probability of voluntarily exiting office. The corresponding change in women's probability of exit is indistinguishable from zero. The insignificant relationships between total hostility and campaigning for a different office indicate that, on average, politicians do not respond to abuse by strategically navigating to a new elective seat.

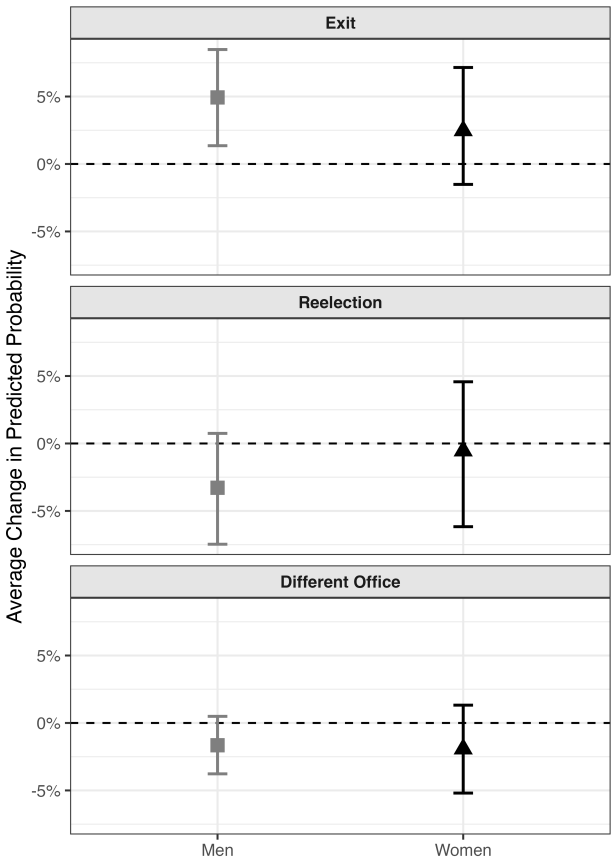
The minimum detectable effect analysis in Appendix D further indicates that the analysis has 80% power to detect a gender difference of approximately 4 percentage points in the

Table 3: The effects of total hostility on campaign choice are concentrated among men

	Model 1	Model 2
Likelihood of Campaign for Reelection vs. Exit		
Visibility	1.074***	1.148***
	(0.024)	(0.036)
Fundraising	1.179**	1.168*
	(0.084)	(0.085)
Democrat	1.278*	1.242*
	(0.129)	(0.130)
Legislative Effectiveness	1.114	1.111
	(0.068)	(0.069)
Nonwhite	1.274	1.284
	(0.167)	(0.168)
Year First Elected	1.273***	1.270***
	(0.051)	(0.052)
Female Political Culture	0.952	0.955
	(0.084)	(0.085)
Woman	1.166	1.100
	(0.131)	(0.144)
Total Hostility		0.269***
		(0.479)
Total Hostility × Woman		1.705
		(0.743)
Likelihood of Campaign for Different Office vs. Exit		
Visibility	1.006	1.117**
	(0.035)	(0.052)
Fundraising	0.933	0.921
	(0.120)	(0.121)
Democrat	0.911	0.874
	(0.200)	(0.202)
Legislative Effectiveness	1.086	1.082
	(0.096)	(0.097)
Nonwhite	1.560*	1.565*
	(0.235)	(0.236)
Year First Elected	1.548***	1.543***
	(0.097)	(0.096)
Female Political Culture	1.113	1.118
	(0.120)	(0.120)
Woman	1.512**	1.428*
	(0.176)	(0.198)
Total Hostility		0.134**
		(0.789)
Total Hostility × Woman		1.608
		(1.063)
Observations	2616	2616
Clusters	1059	1059
Log Likelihood	-1807.483	-1803.050
AIC	3650.966	3650.100

Notes: Relative risk ratios are reported with clustered bootstrap standard errors in parentheses. Standard errors are based on 10,000 bootstrap replications clustered at the individual level. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Figure 1: Increased total hostility corresponds to increased exit for men, but not women



Note: The y-axis indicates the average change in the predicted probability of political exit associated with a 25 percentage point increase in total online hostility. This increment approximates a shift from the 25th to the 75th percentile of total hostility among legislators exposed to hostility. Error bars indicate 95% confidence intervals calculated using 10,000 cluster-bootstrap replications, with legislators as the resampling unit to account for repeated observations of the same individual. Predicted probabilities are calculated from the model reported in Table 3.

effect of an interquartile-range increase in hostility on the probability of exit. The gender differences reported in prior studies of how hostility may affect political ambition are rarely smaller than 4 percentage points (see Table 17) (Håkansson 2024; Håkansson and Josefsson 2026). Thus, while the results cannot rule out smaller gender differences, the analysis is sufficiently powered to detect effects comparable to those typically reported in the existing literature. Women in the dataset are not disproportionately exiting office because of hostility.

Other predictors operate as expected. Visibility and more recent entry into elective office increase the likelihood of both static and progressive ambition relative to exit. Fundraising in the previous cycle is strongly and significantly related to the likelihood of campaigning for a different office. Politicians from underrepresented ethnic groups are significantly more likely than white legislators to pursue a different elected office rather than exit politics. Holding

other covariates constant, the relative risk of seeking another office rather than exiting is approximately 73% higher for nonwhite legislators ($RRR = 1.73, p < .01$). Legislative effectiveness and party identification are not significant. Among legislators who receive no hostility, women are more likely than men to seek a different office rather than exit electoral politics ($RRR = 1.45, p < 0.05$).

Equivalence tests suggest women are less likely to exit

Failure to reject the null for **H1**, however, does not establish that the effects of total hostility on exit are the same for men and women. The confidence interval could still contain fairly large gender differences. Rainey (2014) argues that negligible effects hypotheses—such as **H2**—require more than a confidence interval which includes zero.[21] Instead, researchers should establish what a meaningful effect would look like, then verify that the 90% confidence interval contains only values consistent with the theoretical argument.

I rely on the relevant literature to identify a meaningful effect size. Table 17 compiles several studies on responses to online hostility and ambition which report a change in probability. Prior work identifies effects of approximately four to six percentage points as substantively important. I therefore use ±5 percentage points as the threshold for a meaningful effect.

Next, I estimate the gender difference in the predicted probability of exit when exposure to total hostility increases from the first to the third quartile. I use 10,000 cluster-bootstrap draws to construct a 90% confidence interval for that gender difference. Then I compare that interval with the equivalence bounds of ±5 percentage points. If the entire 90% confidence interval is between -5 and +5 percentage points, I can reject gender differences of a substantively meaningful magnitude. If the confidence interval extends beyond either ±5 boundary, I cannot establish equivalent association between hostility and exit for men and women. Importantly, failing to establish equivalence does not mean that there is a meaningful gender difference, only that I cannot rule one out. The results are displayed in Table 4.

Table 4 fails to establish equivalence between men and women for the exit and reelection outcomes. The estimated effect of increased exposure to total hostility on exit is 2.33 percentage points less for women than for men, suggesting the effect of hostility on the probability of exit is *smaller* for women than men. The confidence interval for campaigning for a different office does remain within the equivalence bounds, implying that there is not a meaningful gender difference in the choice to seek a new elective seat.

The results support the expectation that women are *not* disproportionately more likely than men to exit in association with hostility (**H2**). Although the equivalence test cannot

[21] A negligible effect hypothesis argues that a variable will have no meaningful effect on the outcome.

Table 4: Equivalence tests support H2

Outcome	Gender Difference	90% CI	Equivalent
Exit	-2.33	[-6.05, 1.39]	No
Reelection	2.59	[-1.92, 7.18]	No
Different Office	-0.26	[-3.05, 2.37]	Yes

Note: The gender difference is the average effect of an interquartile-range increase in hostility among women minus the average effect among men. Confidence intervals are 90% percentile intervals from 10,000 bootstrap replications clustered by legislator. Equivalence is established when the full confidence interval for the gender difference falls within ±5 percentage points.

rule out a substantively meaningful gender difference, the uncertainty producing this result is concentrated in the opposite direction from that rejected by **H2**. While I cannot rule out the possibility that online hostility may have a substantially smaller effect among women than men, I can conclude that it does not disproportionately influence women toward exit.

Does the content of hostility matter?

Thus far, the results indicate that exposure to *total* hostility does not disproportionately increase women politicians' likelihood of voluntary exit relative to men. Total hostility, however, does not account for the unique forms of hostility women politicians encounter (Bardall et al. 2020; Jarman 2026). Women in politics receive incommensurate gender-based hostility online, including higher rates of sexualized harassment, gender-based hate speech, and sexist abuse (Bardall et al. 2020; Bjarnegård 2018; Erikson et al. 2023; Esposito and Breeze 2022; Gorrell et al. 2020; Herrick et al. 2022; Holm et al. 2024; Jarman 2026; Kosiara-Pedersen 2023; Southern and Harmer 2019). As a result, we may expect women and men to react differently to gendered than to general hostility.

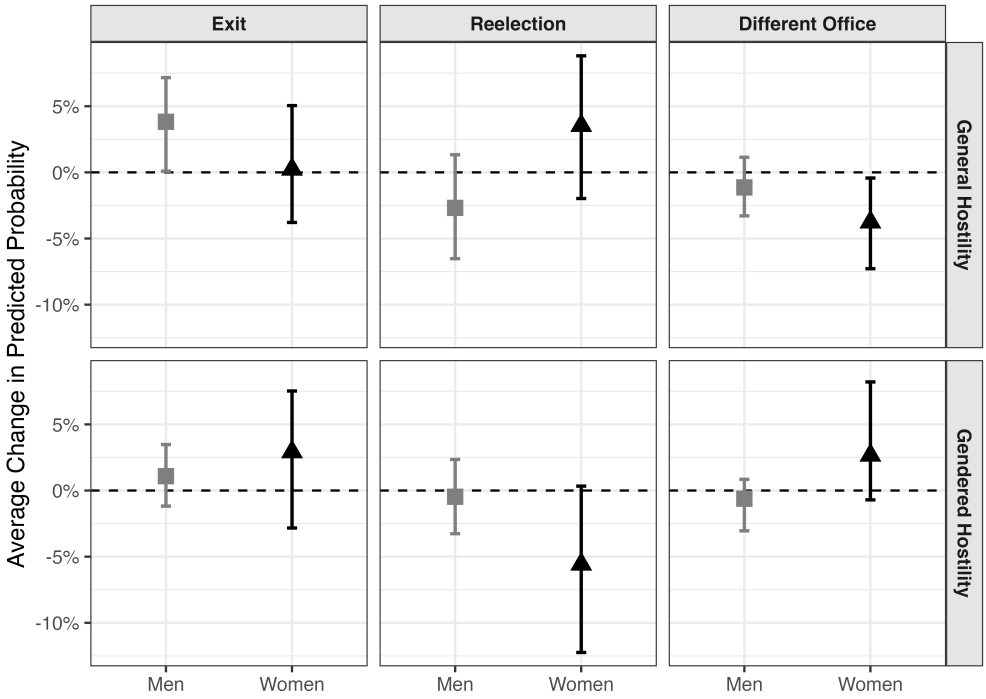
I further test **H1** by separately examining the association of gendered and general hostility with campaign choice. The labeling process scored all public mentions for whether they included gendered content.[22] I can thus create distinct measures of how much *general* hostility and *gendered* hostility politicians received. I replace total hostility in the multinomial model with these two measures and interact both with gender. The resulting regression Table 13 is in the Appendix.

Distinguishing between general and gendered hostility reinforces the prior finding that greater exposure is associated with increased probability of exit among men but not women. Neither interaction is statistically significant, though general hostility is associated with

[22] The full labeling criteria are in Appendix A.

increased likelihood of exit for men. Estimates for gendered hostility are positive for both men and women, however, they are considerably less precise and cannot be distinguished from zero.[23] Overall, the results suggest that the relationship between online hostility and men's political exit is primarily driven by general hostility, while the consequences of gendered hostility remain uncertain.

Figure 2: Men respond to general hostility, women respond to neither



Note: The y-axis indicates the average change in the predicted probability of campaign decision associated with an approximately interquartile-range increase in online hostility. The top row of panels presents general hostility, for which estimates correspond to a 20-percentage-point increase, while the bottom row presents gendered hostility, for which estimates correspond to a 5-percentage-point increase. These increments approximate a shift from the 25th to the 75th percentile of each hostility measure among legislators exposed to that form of hostility. Error bars indicate 95% confidence intervals calculated using 10,000 cluster-bootstrap replications, with legislators as the resampling unit to account for repeated observations of the same individual. Predicted probabilities are calculated from the models reported in Table 13.

In Figure 2, I calculate each politician's predicted probability of exit at their observed levels of general and gendered hostility and after increasing these measures by an interquartile increment. An interquartile increase in general hostility is associated with a four percentage point increase in men's predicted probability of exiting electoral politics. In comparison, greater exposure to general hostility is not associated with any changes in women's campaign

[23] Approximately 19% of men and 20% of women receive at least some gendered hostility, corresponding to 610 and 276 observations, respectively.

decisions. There is not evidence that women are more likely than men to voluntarily exit politics because of gendered hostility.

Figure 11 in the Appendix presents power analyses for the separate measures of general and gendered hostility. The analysis reaches 80% power to detect a gender difference of approximately 10.1 percentage points in the effect of general hostility on the probability of exit and 8.6 percentage points for gendered hostility. These estimates reflect a real boundary in data collection, particularly given the dismantling of the academic Twitter API (Harden and Porter 2026; King and Zeng 2001). As a result, the split-hostility models are sufficiently powered to detect relatively large gender differences, but smaller differences may go undetected. The null interactions in these models should therefore be interpreted more cautiously than the primary Total Hostility analysis in Table 3, for which the minimum detectable gender difference is approximately 4 percentage points (see Figure 10).

Does the severity of hostility matter?

I further test **H1** by considering whether men's association between exposure and exit is explained by a gendered difference in the severity of abuse that men and women receive. I utilize Detoxify to test this possibility. As noted earlier, Detoxify scores each mention on the probability it contains overall toxicity, insults, obscenities, identity attacks, sexually explicit content, and threats. Adding these creates a score indicating the probability that a mention is abusive. Higher scores demonstrate more severe abuse.

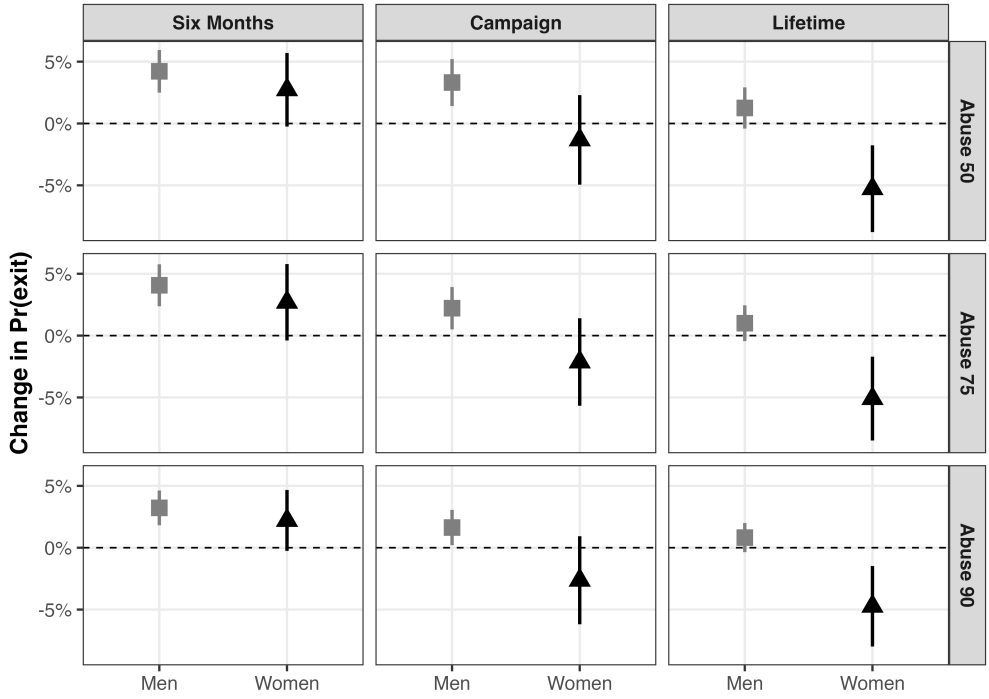
Pairing Detoxify with the BERT labels, I produce three total hostility variables at different thresholds of abuse: 50, 75, and 90. Abuse 50 indicates the proportion of a politician's mentions which were hostile—either general or gendered—and had greater than a 50% chance of being abusive. Abuse 75 and Abuse 90 do likewise for their respective thresholds.

By examining the decision to exit across these thresholds, I ensure that I am comparing men and women who receive approximate severity of abuse. Severity may matter most when received closer to the filing deadline, thus, I generate these variables across three time spans: the six months prior to the filing deadline, the preceding campaign and legislative cycle, and the accumulative exposure since collection began in 2015 to the current filing date. For each of the nine specifications, I estimate the same multinomial logistic regression as seen in Table 3, altering only the measure of hostility interacted with gender. The resulting regression tables are presented in Appendix 4. Figure 3 shows the change in predicted probability associated with an interquartile increase in hostility for each specification.

Even at increasing levels of severity of abuse, the association between online hostility and voluntary political exit is concentrated among men. As hostility becomes more severe,

the RRRs generally become smaller, indicating that increased severity of abuse produces a stronger estimated association between hostility and exit. There is no evidence, however, that women become disproportionately more likely to exit as the severity threshold rises. For each threshold of abuse within the six month and prior campaign windows, the estimated gender interaction is statistically indistinguishable from zero.

Figure 3: The pattern persists across severity thresholds and time windows



Note: The y-axis indicates the average change in the predicted probability of campaign decision associated with an approximately interquartile-range increase in online hostility. Columns distinguish the window over which hostility is measured: the six months preceding the filing deadline, the campaign period, and cumulative exposure since data collection began in 2015 ("Lifetime"). Rows distinguish the threshold used to classify abusive tweets using Detoxify: Abuse 50, Abuse 75, and Abuse 90 indicate predicted abuse probabilities of at least 0.50, 0.75, and 0.90, respectively. Error bars indicate 95% confidence intervals calculated using 10,000 cluster-bootstrap replications, with legislators as the resampling unit to account for repeated observations of the same individual. Predicted probabilities are calculated from the models reported in Appendix 4.

Increased *accumulative* exposure to hostility up to that filing deadline does produce a significant difference between men and women. Crucially, the estimated interaction signifies that hostility is associated with a significantly weaker tendency toward exit among women than among men. Because this measure captures hostility accumulated from the beginning of data collection in 2015 through each politician's filing deadline, however, the result may also capture selection into prolonged exposure. This possibility raises an important question for interpreting the relationship between hostility and political ambition: are the politicians

most likely to remain and advance in politics also those who receive the most hostility?

Do ambitious politicians receive more hostility?

Thus far, the results indicate that the depressing effects of online hostility on political ambition are concentrated among men. At the same time, there are many politicians whose ambition does not appear to be influenced by online hostility, despite encountering high levels of exposure. For these individuals, it is necessary to identify whether political ambition may condition how much hostility they receive. Exposure to higher levels of hostility appears to affect some politicians' ambition, but do ambitious individuals also receive higher levels of online hostility?

More visible individuals both receive greater proportions of hostility in their mentions and are less likely to exit. This may reflect a two-step process where online users are not only able to successfully identify politicians with greater static or progressive ambition, but also choose to target those politicians with more hostility. Under such a mechanism, some of the legislators who receive high proportions of abuse would also be the legislators most committed to pursuing their political career. If this pattern is present in the data, it will result in more conservative estimates of the negative effect of hostility on ambition. I assess whether ambitious politicians receive more hostility in three analyses.

First, I examine whether legislators who later exited politics exhibited systematically different patterns of Twitter behavior before the filing deadline. If ambitious politicians are readily identifiable from how they communicate online, legislators who remained in office should differ from those who exited across observable features of their Twitter activity. I use data created by Butler et al. (2023) to estimate whether the politician's Twitter behavior—tweet volume, ideology, sentiment, opinion expression, fundraising appeals, and policy content—distinguish legislators who subsequently exit from those who remain. None of the Twitter behaviors is individually associated with subsequent exit, and the measures are not jointly associated with exit ($\chi^2(6) = 8.34$, $p = .214$) (see Tables 18 and 19 in the appendix). While this analysis cannot rule out the possibility that ambitious politicians communicate differently online, it does not provide evidence that Twitter users can infer ambition from these observables.

Second, I investigate whether hostile users disproportionately target high-potential politicians. I consider conventional indicators of quality: legislative effectiveness, tenure in office, and lagged fundraising and donor count. In the previous models, I included these quality indicators as controls. Now, I use them as the primary explanatory variables for how much

total hostility an individual received in the six months preceding a filing deadline.[24] In addition, I control for gender, partisanship, ethnicity, and women-friendly culture.[25] Results are displayed in Table 20. The results do not suggest that hostile users are disproportionately targeting politicians based on these indicators of quality. Politicians who are more effective in legislatures and had higher fundraising in previous campaign cycles do receive *less* total hostility; however, these differences are not statistically significant at conventional levels.

Lastly, I consider whether online users direct more hostility toward ambitious politicians by identifying whether legislators who announce another campaign see a larger increase in hostility after the filing deadline than politicians who voluntarily exit.[26] To test this, I regress the proportion of hostility in the six months following the filing deadline on the proportion of hostility in the six months preceding it.[27] The key independent variable in the analysis of covariance (ANCOVA) model is campaign choice, with exit serving as the reference category.[28] Seeking reelection is associated with approximately a four percentage point increase in the share of hostile tweets relative to exiting politics. Politicians pursuing other offices are not statistically distinguishable from those who exit. Table 21 and Figure 12 present these results.

Taken together, these three analyses provide sparse evidence that the politicians most likely to remain and advance in politics are also those who receive the most hostility. Hostile users do not seem to be successfully identifying or targeting politicians with greater ambition based on their Twitter behavior or conventional indicators of quality. Thus, I conclude that while users may respond to the act of remaining in politics, the reception of online abuse does not itself appear to be a proxy for political ambition. This finding further bolsters the rejection of **H1** and support for **H2** based on the results thus far.

[24] I model outcomes using linear regression because the dependent variables are continuous proportions $y \in (0, 1]$.

[25] I limit the sample to active politicians who received at least one hostile tweet. Politicians who are not currently in office receive far fewer mentions, which inflates the relative proportion of mentions they receive that are hostile. Additionally, if they have been out of office for multiple cycles, they lack the relevant fundraising measures to approximate quality.

[26] For candidates who announce their candidacy before the filing deadline, the filing deadline can still mark an increase in campaign activity and corresponding public awareness.

[27] I restrict the sample to legislators who are in office and received more than ten tweets during each six month period.

[28] All models control for hostility during the six months preceding the filing deadline, legislator visibility, fundraising, gender, race, party, and year fixed effects. Standard errors are clustered by legislator.

Part 2: Is this a gender-based selection effect?

The expectation that exposure to hostility would not disproportionately increase women politicians' likelihood of voluntary exit relative to men (**H2**) arose from gender-based selection theory. Taking this application further, **H3** posits that the more gendered barriers women faced in entering office, the less likely they will be to voluntarily exit politics in relation to online hostility. Essentially, if women's demonstrated resilience is the result of a gender-based selection effect, women who faced particular difficulty in entering political office will be the least reactive to online abuse.

States which have historically elected higher levels of women to public office may be more likely to have a culture conducive to women playing a role in politics (Ladam et al. 2018; Windett 2011). As a result, women politicians from states with more women-friendly political cultures may have faced fewer gendered barriers to office-holding.[29] Ladam et al. (2018) builds off Windett (2011)'s similar measure, retaining indicators for the percentage of women in statewide elected executives and members of the U.S. Congressional delegation. Importantly for my dependent variable, Ladam et al. (2018) omit the percentage of women state senators and lower chamber representatives. Ladam et al. (2018) also includes three indicators of equality in society based on educational attainment, work force inclusion and general feelings toward statewide equality.[30]

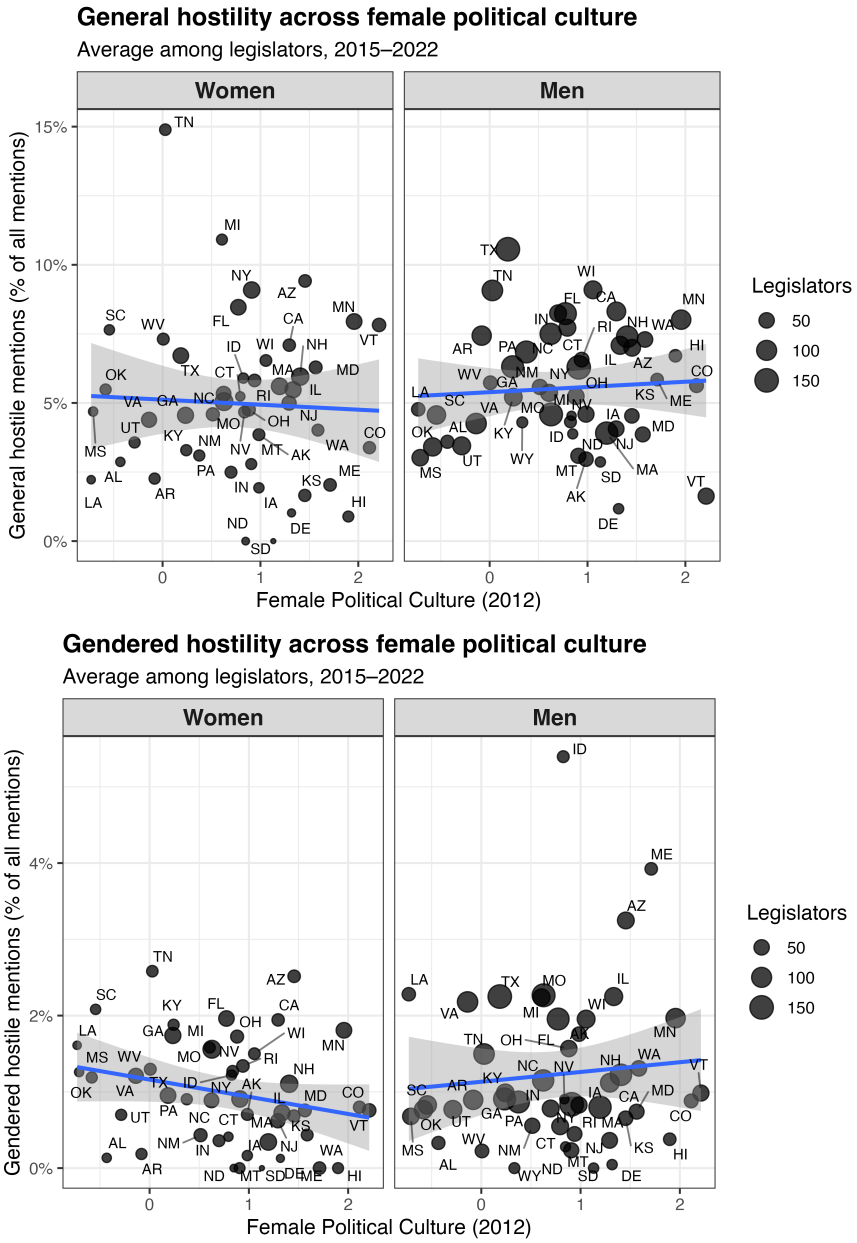
Figure 4 maps the relationship between how women-friendly each state's political culture is and the general and gendered hostility men and women receive in those states. There is no clear pattern for general hostility. However, there does appear to be some evidence that as states become more friendly toward women in office, those women receive a smaller proportion of gendered hostility in their mentions. This result is consistent with the pattern identified in Jarman (2026), where the growth rate of women in state legislatures correlates to a gendered difference in the content of abuse but not the overall volume of hostility. The weak relationship between state culture and hostility suggests that women are not exposed to dramatically more abuse in low-culture states, building confidence in the appropriateness of the following analysis.

Among women legislators, I examine whether the relationship between online hostility and campaign choice varies by how women-friendly their state political culture is. I estimate a multinomial logit model that interacts the total hostility a legislator receives with Ladam et al. (2018)'s measure of women political culture. I then calculate how an interquartile

[29] In tables and figures, I refer to Ladam et al. (2018)'s measure as they do: "female political culture". In discussing these results, I opt for the more inclusive term "women-friendly political culture".

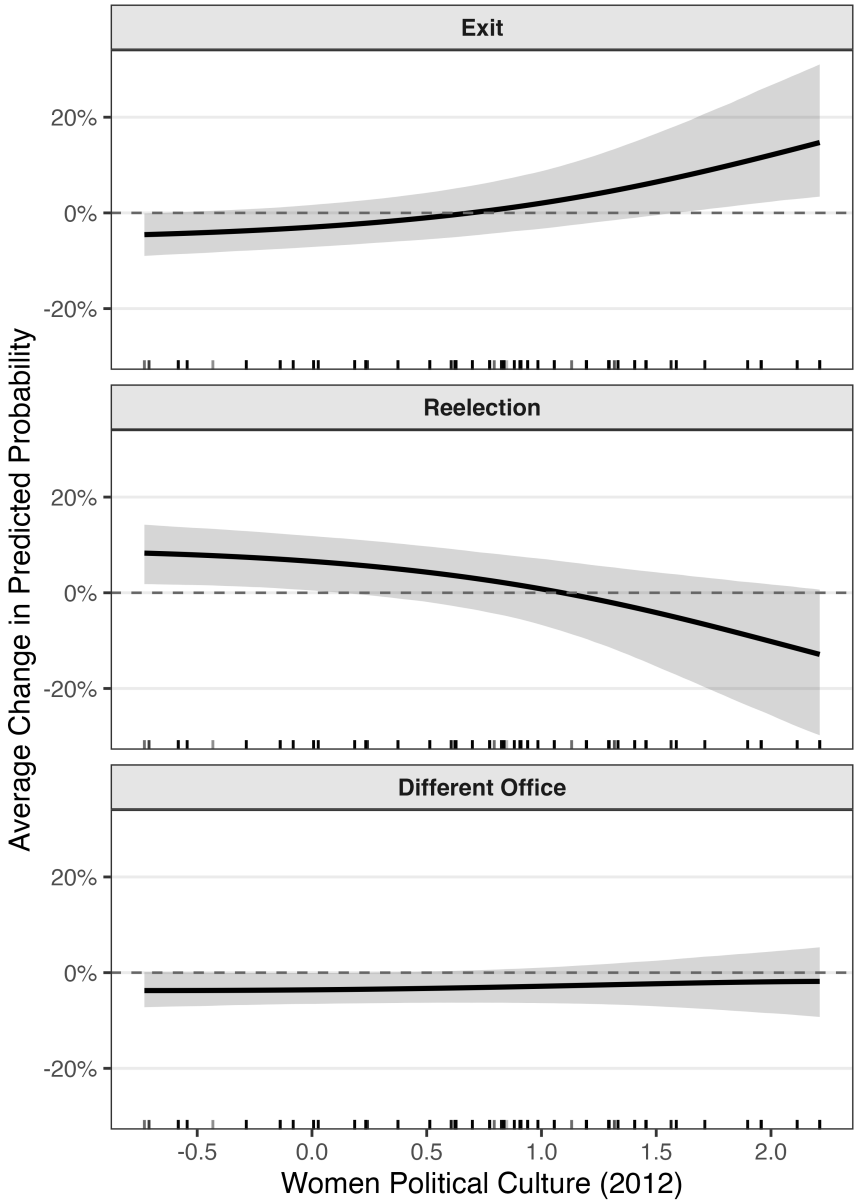
[30] To learn more about this measure, view Ladam et al. (2018)'s supporting information on https://doi.org/10.1111/ajps.12351.

Figure 4: Mapping the relationship between online hostility and female political culture



Note: The x-axis indicates how friendly a state's political culture is toward women, with higher values indicating a more women-friendly culture. The y-axis in the top figure is the percentage of the mentions which politicians receive with general hostility in the six months prior to the filing deadline for their state. The y-axis in the bottom figure is a similar percentage, but for mentions with gendered hostility. Both plots are split into separate panels for men and women legislators in their state.

Figure 5: Total hostility has less effect in political cultures which are not women-friendly



Note: The x-axis indicates how friendly a state's political culture is toward women, with higher values indicating a more women-friendly culture. The y-axis shows the average change in the predicted probability of each campaign choice associated with a 25-percentage-point increase in total hostility. Shaded areas indicate two-sided 90% confidence intervals calculated using 10,000 bootstrap samples clustered by legislator. Predicted values come from Table 22.

increase in total hostility changes the predicted probability that a woman campaigns for reelection, runs for a different office, or voluntarily exits across the observed range of women-friendly political cultures. I use ten-thousand bootstrap samples clustered by legislator to calculate uncertainty around these estimates. Figure 5 displays the results.

A statistically significant association exists between how women-friendly a state's political culture is and the effect of total hostility on campaign choice among its women politicians. In the most women-friendly states, an interquartile increase in hostility is associated with a substantially higher probability of exit (15%) and a correspondingly lower probability of seeking reelection. This implies that women who faced greater barriers in entering political office are less likely to be influenced by total hostility, providing strong support for **H3**. The probability of campaigning for a different office changes little across women-friendly political cultures. The estimates are less precise at the extremes of the political culture measure, where observations are scarcer. The relationship between total hostility and exit strengthens in states with more women-friendly political cultures. Gender-based selection theory is an important lens when considering whether politicians voluntarily exit office because of online hostility.

Discussion and Conclusion

Dealing with trolls and online hostility in general has become an unavoidable component of being a politician (Lai et al. 2022; Morton et al. 2023; Ramachandran et al. 2024; Simi et al. 2024). Scholars and political operatives alike have speculated that this relatively new cost of a political career may influence some politicians to exit elective office voluntarily (Frankel 2020; Jankowicz 2022; Phillips 2017; Sobieraj 2020). In particular, the dialogue on how online hostility is reshaping who enters, exits, and stays in office has predominantly focused on the consequences for women's descriptive representation (Carson et al. 2024; Erikson et al. 2023; Harmer and Southern 2023; Herrick and Franklin 2019; Herrick and Thomas 2022; Herrick et al. 2026; Håkansson 2024; Håkansson and Lajevardi 2024; Jankowicz 2017; Jarman 2026; Kosiara-Pedersen 2023; Krook 2017, 2020; Sobieraj 2018; Thomas et al. 2019; Yan and Bernhard 2024).

This paper is the first to identify whether online hostility is driving incumbent politicians from office. Using nearly a decade of observational panel data on the campaign choices and online environment of 1,247 US politicians, I estimate whether exposure to online hostility influenced subsequent decisions to campaign for reelection, seek a different office, or voluntarily exit. The results offer three contributions to the literature and point to worthwhile directions for future research.

The first contribution is that online hostility toward politicians is a severe form of political violence that influences who stays in office and who exits. Second, contrary to common expectations, the findings indicate that the effects of online hostility on campaign choice are concentrated among men—not women—in politics. Increased exposure to hostility corresponds with an approximately 5 percentage point increase in men's likelihood of voluntarily exiting politics, but has no discernible effect on women. This pattern persists across a plethora of alternative specifications and robustness checks. Women incumbents are not disproportionately exiting politics due to online hostility. Third, women politicians who faced more gendered barriers in entering office are the least likely to voluntarily exit in relation to online hostility, indicating that gender-based selection does influence the response to abuse.

Future research should wade into the underlying mechanisms—both psychological and material—which make online hostility a challenge when campaigning for reelection or another office. Sifting through deluges of abuse to identify genuine constituent communication requires additional time and emotional bandwidth (Schriock and Reynolds 2021; Sobieraj 2020). Constituent service is similarly restrained when fungible resources are instead devoted toward categorizing and reporting credible threats to platforms and law enforcement (Citron 2014). Further, the commonly reported behavioral changes following exposure, such as avoiding media appearances and public events, are antithetical to the requirements of a successful campaign (Ramachandran et al. 2024). It is additionally important to create practical implementations to mitigate the costs of online hostility. The rise of image-based abuse—which this paper does not account for—heightens the urgency of developing solutions to protect politicians.

The impacts of online hostility on function and ambition are likely greater for politicians with fewer resources to screen abuse. Local and state officials have far fewer staff than national politicians and are more likely to operate their own social media accounts, increasing the likelihood of direct contact with abuse (Butler et al. 2023; Wood 2021). It is important for future research to both compare the consequences of online hostility for politicians at different levels of government and engage in understanding the personality traits of those who persevere through abuse to remain in politics. Only individuals who can tolerate online hostility at local and state levels will progress through the pipeline to national office, which has historically relied on former state legislators in particular (Holman 2017; Manning 2024; McCrain and O'Connell 2023). As a result, online hostility may alter the composition of politicians at all levels of government.

Women's resilience opens new questions about the conditions under which women remain in politics despite sustained abuse. Likely, women's resilience to the maltreatment they experience as a public figure is dependent on individual traits and circumstances. Personality

characteristics, domestic status, the severity of the offense, and structural supports all feasibly influence retention and exit. Future research on violence against women in politics—online and offline—should attempt to disentangle the causal mechanisms behind different responses to hostility within and across demographic groups.

An important implication from the findings in this paper is that if online hostility disproportionately reduces women's descriptive representation, it likely does so by discouraging entry rather than driving exit. Future research should consider whether online hostility functions similarly to other demonstrated costs of office in winnowing the pool of future candidates. It is essential to determine whether online hostility is disproportionately dissuading certain demographics from political participation and ambition.

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Supplementary Material for: Toll of Trolls: Are Politicians Voluntarily Exiting Office Because of Online Hostility? Annie JarmanPostdoctoral Research Associate, University of Notre Dame. Email: ejarman@nd.edu. Web: https://anniejarman.github.io.

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A Labeling Mentions

1 Instructions for *Hostile* label

Hostility Classification: Label tweet 'hostile' or 'not hostile'. A tweet is 'hostile' if it does ANY of the following:

- Is strongly critical of @legislator
- Uses impolite, rude, or hurtful language
- Uses derogatory comments, swear words, vulgarities, threats, or hate speech
- Uses sarcasm or mocking tones, e.g., thanks a lot
- Has more than one question mark or an ellipsis '...'
- Uses dismissive or condescending language, including diminutives like 'Sweetie' or 'Honey', or suggests a statement by @legislator isn't important
- Includes sexual objectification or harassment
- Manifests racism, misogyny, homophobia, or attacks based on politics, religion, etc
- Uses ad hominem or personal attacks, such as accusing @legislator of having vices, being ineffective, corrupt, bigoted, etc
- Demands @legislator resign, belittles politicians, or advocates for electoral loss
- Cannot be confidently labeled as not hostile
- Includes hostile hashtags or emojis. Ignore URLs

2 Instructions for *Gendered* label

Gendered Classification: Label tweet 'gendered' or 'not gendered'. A tweet is 'gendered' if it does ANY of the following:"

- Contains the specific words 'he', 'his', 'him', 'she', 'her', 'hers'. Pay special attention to these words
- Comments on @legislator's physical appearance, including wardrobe or pictures
- References @legislator's romantic or sexual relationships, marital or parental status, e.g., calling them a mother or father
- Makes judgments about competence based on @legislator's gender
- References stereotyped masculinity or femininity

- Mentions @legislator's gender explicitly, like stating they are a man, woman, congressman, or congresswoman
- Encourages the election of a specific gender
- Compares @legislator to an individual of the opposite gender
- Uses gendered slurs or slang like 'bitch', 'dick', 'asshole', 'dude', 'yes man', or 'girl boss'
- Cannot be confidently labeled as not gendered
- Includes gendered hashtags and emojis. Ignore URLs

3 Instructions for *Supportive* label

"*Supportive Classification*: Label tweet 'supportive' or 'not supportive'." A tweet is 'supportive' if it does ANY of the following:"

- Any 'thanks' (thanks/TY) in any context.
- Explicit support for the campaign (voting, donating, volunteering).
- Encouraging others to pray for @legislator or highlighting @legislator as a minority.
- Praise for legislative, public, or constituency work.
- Positive comparisons of @legislator to others.
- Compliments (bill passage, milestones, birthdays, sympathy).
- Friendly relationships or concern for @legislator's well-being.
- Positive interactions ('glad to be with')
- Vague niceties ('super', 'nice'), third-party updates, or cooperative actions without praise are not supportive.
- Hashtags/emojis can indicate support. Ignore URLs.

4 Partitioning training data

To build the training data, I randomly sampled tweets from each legislator to create a training corpus of 152,522 mentions, equal to 1% of the entire corpus. I partitioned the scoring subset into four groups, described in Table 1 (Park and Montgomery 2025). The first three sets were all labeled using gpt-4-turbo zero-shot classification on the OpenAI API, described below. Groups B and D were hand-labeled by expert coders to validate the zero-shot classification labels and final measure, respectively. Groups A and B are the training data, whereas Groups C and D are held out from BERT finetuning to validate the final measure.

Table 5: Partitioning training corpus for classification and validation

	Group A (n=122,615)	**Group B** (n=2,991)	**Group C** (n=23,925)	**Group D** (n=2,991)
Zero-shot classified label	✓	✓	✓	
Expert label		✓		✓
Held out from model training			✓	✓

5 Feature representation and model validation

Table 6: Zero-shot predictions (before DSL) vs. group B

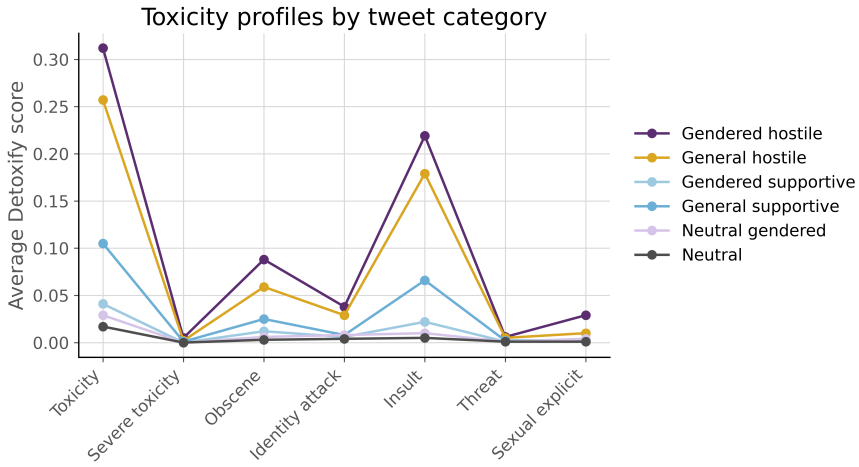
Metric	**Gender**	**Hostile**	**Supportive**
Accuracy	0.9364	0.8929	0.9237
Recall	0.6256	0.8272	0.8610
Precision	0.7651	0.7558	0.7931
F1 score	0.6883	0.7899	0.8256
ROC_AUC	0.8007	0.8706	0.9007

Table 7: BERT predictions (before DSL) vs. groups B & D

Metric	**Gender**	**Hostile**	**Supportive**
Accuracy	0.9718	0.8861	0.9265
Recall	0.9113	0.8295	0.8850
Precision	0.8486	0.7362	0.7900
F1 score	0.8789	0.7801	0.8348
ROC_AUC	0.9454	0.8669	0.9113

6 Comparison of BERT-derived labels with Detoxify

Figure 6: Detoxify validates BERT-Hostile label

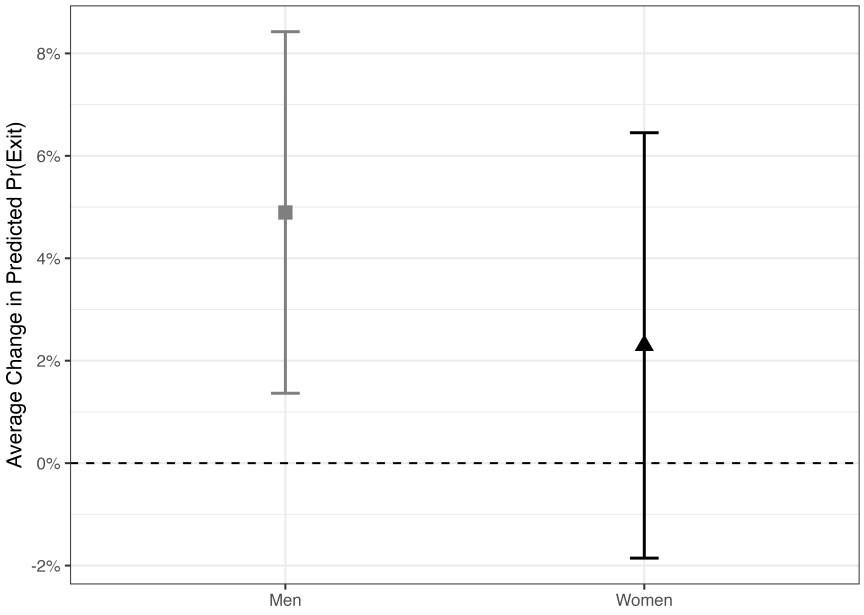


B Alternate Model Specifications for Effect of Hostility on Exit

1 Binomial logit model

This paper is primarily framed around retention rather than trajectory. As a result, I include a variation of Table 3 which utilizes a binary dependent variable for whether a politician exited or not. Accordingly, I use a binomial logit instead of a multinomial logit. I find the same core result where the association between hostility and exit is concentrated among men.

Figure 7: Total hostility predicts political exit among men, but not women



Note: The y-axis indicates the average change in the predicted probability of political exit associated with a 25-percentage-point increase in total online hostility. This increment approximates a shift from the 25th to the 75th percentile of total hostility among legislators exposed to hostility. Error bars indicate 95% confidence intervals calculated using standard errors clustered by legislator. Predicted probabilities are calculated from the model reported in Table 8.

Table 8: Results for Table 3 similar as binomial logit

	Model 1	Model 2
Intercept	0.27***	0.26***
	(0.12)	(0.12)
Log Mentions	0.93***	0.87***
	(0.02)	(0.03)
Fundraising	0.85*	0.86*
	(0.08)	(0.08)
Democrat	0.80*	0.82
	(0.13)	(0.13)
Legislative Effectiveness	0.90*	0.90*
	(0.06)	(0.06)
Nonwhite	0.77	0.77*
	(0.16)	(0.16)
Year First Elected	0.77***	0.78***
	(0.05)	(0.05)
Female Political Culture	1.02	1.02
	(0.08)	(0.08)
Woman	0.85	0.90
	(0.13)	(0.14)
Total Hostility		3.86***
		(0.45)
Total Hostility × Woman		0.57
		(0.68)
AIC	2127.9	2124.2
Observations	2566	2566
Clusters	1056	1056

Odds ratios reported. Cluster-robust standard errors clustered by legislator in parentheses. Inactive legislators, federal office seekers, term-limited legislators, and appointed officials are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

2 Distinguishing between higher and lower offices

Theoretically, we may expect politicians to respond to hostility by campaigning for either "higher" or "lower" offices. An individual's place in the pipeline strongly correlates with how visible they are to the public, which in turn influences how much online hostility they receive (Bjørgo 2022; Håkansson 2021; Jarman 2026; Rheault et al. 2019; Theocharis et al. 2020). A politician who wished to reduce their exposure to online abuse could choose to move to a lower profile office. There is some support for this in the literature. In a survey of politicians, women indicated twice as much willingness as men to decrease their pay in exchange for reduced harassment (Kjøller and Pedersen 2025).

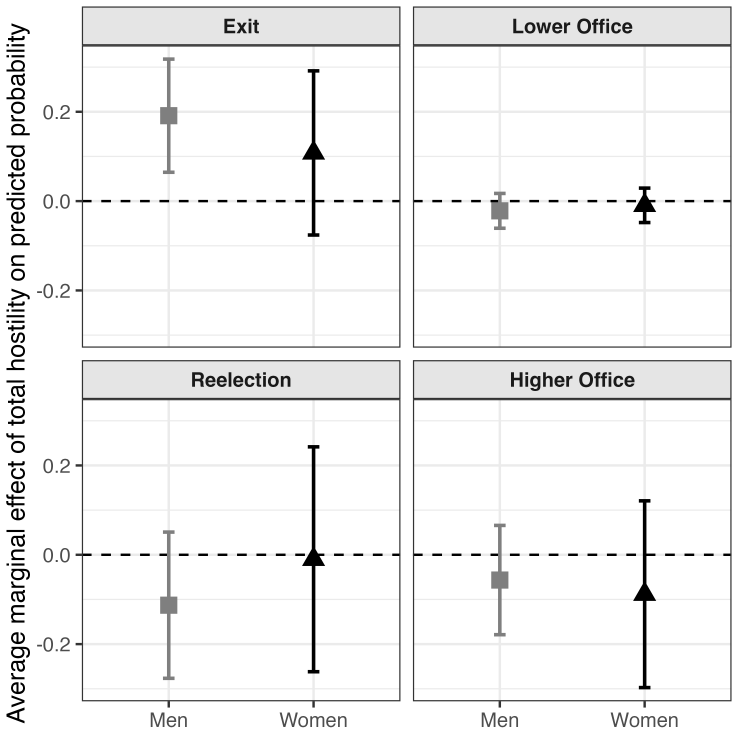
Alternatively, some women respond to hostility with increased ambition. In some cases, hostility can be politically mobilizing for women because staying out of politics becomes perceived as too politically costly (Agerberg and Kreft 2023; Clayton et al. 2023; Dittmar 2020). Mobilization in response to marginalization is most pronounced among individuals with experiences of discrimination, intimating that women politicians with higher exposure to hostility will be more likely to campaign for more influential offices (Kjøller and Pedersen 2025; Pedersen et al. 2025).

To find out, I rank offices as higher or lower by considering whether the seat has more power within that same state's context. For instance, national positions and statewide executive seats are considered higher than a state legislative seat, whereas local or county level roles are considered lower. A full documentation of which offices appear in the dataset, at what frequency, and how they are coded are available in Table 9.

I repeat the multinomial model, this time with separate categories for movement toward a higher or lower office. The choice to not campaign is the reference category. Most individuals move toward seats with more power rather than less, creating stable estimates for the decision to campaign for reelection or higher offices rather than exit, but imprecise results for the choice to campaign for lower office. As a result, the estimates for lower-office candidacy should be interpreted with caution as suggestive rather than definitive. Table 10 displays the results.

Distinguishing between higher and lower offices within the choice to campaign for a different office does not alter the results found in the main text. Men continue to be more likely than women to exit office in correlation with greater exposure to total hostility. Exposure to total hostility does not appear to cause either higher or lower offices to appear more attractive than the current office.

Figure 8: General hostility redirects men from reelection but not higher or lower offices



Note: The y-axis indicates the average marginal effect of general online hostility on the probability of the outcome listed at the top of each panel. Predicted values come from Table 10.

Table 9: Offices by Gender and Power Assessment

Office	**Men**	**Women**	**Total**	**Likely to be More Powerful?**
Governor	5	4	9	Yes
Lieutenant Governor	7	4	11	No (most states ceremonial)
State Senator	99	51	150	Yes
U.S. Senator	3	2	5	Yes
U.S. Representative	21	16	37	Yes
Secretary of State	5	4	9	Yes
Attorney General	4	1	5	Yes
State Auditor / Comptroller (incl. Missouri State Auditor, Auditor, Comptroller)	3	3	6	Yes
State Treasurer	1	2	3	Yes
Commissioner of Insurance	1	0	1	Yes
Superintendent of Public Instruction	1	0	1	Yes
State Supreme Court Justice (incl. "State Supreme Court")	1	1	2	Yes
Public Service Commission (incl. Montana exception)	2	2	4	No (except MT)*
Court of Appeals (state)	1	0	1	No
Trial Court Judge (civil, superior, county circuit)	2	4	6	No
Mayor	1	1	2	No (except big cities)
County Commission	4	1	5	No
City Council / Alderman	3	1	4	No
County Legislature	1	0	1	No
County Sheriff	1	0	1	No
District Attorney (U.S. Attorney if federal)	1	0	1	Yes
Local School / Education Board (incl. school boards, community college boards)	1	1	2	No
Local Financial / Administrative Officers (incl. county auditors, municipal treasurers, property appraisers)	0	3	3	No
Federal Agency State Director (incl. USDA, Farm Service Agency roles)	0	3	3	Yes
U.S. Ambassador	1	0	1	Yes
Deputy or Assistant to Elected Official (incl. Deputy Secretaries, Borough Presidents, Chiefs of Staff, Policy Directors, Budget Commissioners, Parole Board, Supervisor of Huntington)	11	3	14	No

Table 10: Distinguishing between higher and lower office doesn't change results

	Lower Office	**Reelection**	**Higher Office**
	vs. Exit	vs. Exit	vs. Exit
Visibility	1.046	1.147***	1.135**
	(2.068)	(0.037)	(0.054)
Fundraising	0.816	1.168*	0.948
	(0.229)	(0.087)	(0.133)
Democrat	1.264	1.242*	0.831
	(0.717)	(0.130)	(0.207)
Legislative Effectiveness	0.770	1.112	1.109
	(0.316)	(0.066)	(0.094)
Nonwhite	4.883***	1.283	1.185
	(0.504)	(0.170)	(0.247)
Year First Elected	1.672	1.270***	1.519***
	(0.363)	(0.049)	(0.098)
Female Political Culture	1.486	0.955	1.071
	(0.271)	(0.084)	(0.128)
Woman	0.849	1.100	1.557**
	(0.477)	(0.151)	(0.202)
Total Hostility	0.005	0.270***	0.158**
	(371.663)	(0.498)	(0.807)
Total Hostility × Woman	13.161	1.706	1.353
	(187.259)	(0.734)	(1.116)
Observations		2616	
Clusters		1059	
Log Likelihood		-1884.815	
AIC		3835.630	

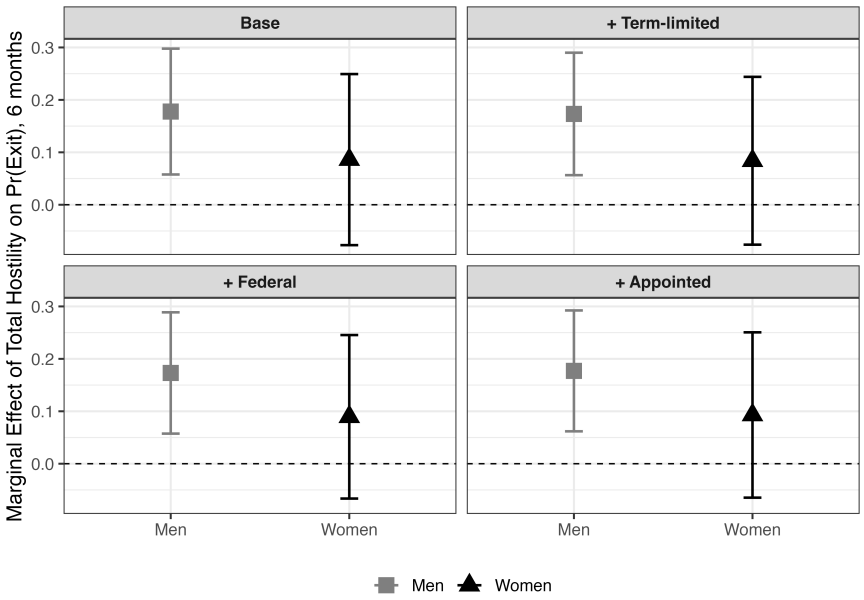
Notes: Relative risk ratios are reported with clustered bootstrap standard errors in parentheses. Standard errors are based on 1,000 bootstrap replications clustered at the individual level. Exit is the reference outcome. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

3 Probability of exit, across sample definitions

I consider four restrictions to the dataset. The 'base' model dataset excludes three potential edge cases. These include legislator-election pairs where the legislator is being term-limited out of office, federal offices, and appointed offices. Model 1 in Table 11 is a multinomial logit for the 'base' sample. Model 2 shows the results for the base sample plus legislators running up against term-limits. This expansion adds seventy-one observed politician-campaign year dyads back to the sample. Model 3 further expands the sample by adding fifty-eight campaign decision observations for politicians in federal elected positions. Lastly, Model 4 contributes an additional twenty-four dyads for individuals who leave an elective position for an appointed one. Whereas Table 3 reports bootstrapped and clustered standard errors, Table 11 reports model-based standard errors from the untransformed coefficient.

Each of these edge case expansions are comparably small next to the base model, which may explain why they have such limited model influence. Both term-limited individuals and appointed individuals are more likely to choose to not campaign. As seen in Figure 9, across each sample definition men continue to display a higher marginal probability of exiting politics due to total hostility than women show. This robustness check solidifies the findings presented in the main text.

Figure 9: Marginal effect of total hostility (6 months) on probability of exit across samples



Note: Predicted values are obtained from Table 11.

In addition, I check the relationship with these same four sample definitions for the proportion of total hostility politicians received in the 30 months prior to the filing deadline. The relative risk ratios are slightly higher for total hostility in the 6-month table than the 30-month table. This may indicate a slight recency bias in how total hostility affects the decision to campaign.

Table 11: Multinomial models across sample definitions, 6 months

	Base	+ Term-Limited	+ Federal	+ Appointed
Likelihood of Campaign for Reelection vs. Exit				
Visibility	1.148*** (0.037)	1.146*** (0.037)	1.146*** (0.037)	1.154*** (0.037)
Fundraising	1.214** (0.080)	1.214** (0.080)	1.176** (0.079)	1.159* (0.079)
Democrat	1.237 (0.131)	1.232 (0.131)	1.219 (0.130)	1.243* (0.130)
Legislative Effectiveness	1.113* (0.064)	1.114* (0.064)	1.112* (0.064)	1.118* (0.064)
Nonwhite	1.289 (0.168)	1.290 (0.167)	1.296 (0.167)	1.250 (0.166)
Year First Elected	1.269*** (0.051)	1.269*** (0.051)	1.267*** (0.051)	1.267*** (0.050)
Total Hostility	0.271*** (0.475)	0.271*** (0.474)	0.269*** (0.475)	0.259*** (0.475)
Woman	1.075 (0.149)	1.070 (0.148)	1.088 (0.148)	1.065 (0.147)
Total Hostility × Woman	1.776 (0.774)	1.790 (0.772)	1.726 (0.770)	1.746 (0.768)
Term-Limited		0.003*** (1.011)	0.003*** (1.011)	0.003*** (1.011)
Appointed				0.023*** (0.758)
Likelihood of Campaign for Different Office vs. Exit				
Visibility	1.174*** (0.060)	1.172*** (0.060)	1.115* (0.059)	1.121* (0.059)
Fundraising	0.886 (0.113)	0.886 (0.113)	0.909 (0.107)	0.899 (0.107)
Democrat	0.873 (0.207)	0.870 (0.207)	0.901 (0.195)	0.934 (0.194)
Legislative Effectiveness	1.114 (0.098)	1.115 (0.098)	1.088 (0.095)	1.093 (0.095)
Nonwhite	1.519* (0.251)	1.519* (0.251)	1.544* (0.236)	1.466 (0.235)
Year First Elected	1.562*** (0.106)	1.562*** (0.106)	1.560*** (0.099)	1.559*** (0.099)
Total Hostility	0.127** (0.913)	0.127** (0.912)	0.135** (0.889)	0.130** (0.889)
Woman	1.345 (0.223)	1.339 (0.223)	1.437* (0.208)	1.392 (0.206)
Total Hostility × Woman	1.990 (1.220)	1.999 (1.219)	1.637 (1.200)	1.718 (1.198)
Term-Limited		0.000 (102.779)	0.000 (118.339)	0.000 (60.886)
Appointed				0.079** (1.041)
Observations	2909	2980	3038	3062
Clusters	1203	1223	1227	1235
Log Likelihood	-1709.35	-1714.78	-1809.70	-1819.96
AIC	3458.70	3473.56	3663.40	3687.92
McFadden R^2	0.126	0.178	0.176	0.187

Notes: Relative risk ratios are reported with standard errors in parentheses. Exit is the reference outcome. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table 12: Multinomial models across sample definitions, 30 months

	Base	+ Term-Limited	+ Federal	+ Appointed
Likelihood of Campaign for Reelection vs. Exit				
Visibility	1.065** (0.030)	1.064** (0.030)	1.064** (0.030)	1.074** (0.030)
Fundraising	1.231*** (0.080)	1.232*** (0.080)	1.196** (0.079)	1.177** (0.079)
Democrat	1.214 (0.132)	1.209 (0.132)	1.197 (0.132)	1.216 (0.132)
Legislative Effectiveness	1.109 (0.064)	1.111 (0.064)	1.108 (0.064)	1.114* (0.064)
Nonwhite	1.286 (0.167)	1.287 (0.167)	1.292 (0.167)	1.246 (0.165)
Year First Elected	1.271*** (0.050)	1.272*** (0.050)	1.270*** (0.050)	1.270*** (0.050)
Total Hostility	0.231*** (0.560)	0.231*** (0.560)	0.227*** (0.561)	0.201*** (0.559)
Woman	1.197 (0.160)	1.191 (0.159)	1.212 (0.159)	1.176 (0.157)
Total Hostility × Woman	0.703 (0.778)	0.714 (0.777)	0.682 (0.775)	0.721 (0.771)
Term-Limited		0.003*** (1.010)	0.003*** (1.010)	0.003*** (1.010)
Appointed				0.020*** (0.759)
Likelihood of Campaign for Different Office vs. Exit				
Visibility	1.038 (0.048)	1.037 (0.048)	0.995 (0.047)	1.001 (0.047)
Fundraising	0.905 (0.113)	0.905 (0.113)	0.929 (0.107)	0.918 (0.107)
Democrat	0.880 (0.209)	0.877 (0.209)	0.901 (0.196)	0.931 (0.196)
Legislative Effectiveness	1.113 (0.098)	1.114 (0.098)	1.087 (0.095)	1.091 (0.095)
Nonwhite	1.521* (0.251)	1.522* (0.251)	1.540* (0.236)	1.460 (0.235)
Year First Elected	1.567*** (0.106)	1.567*** (0.106)	1.564*** (0.099)	1.564*** (0.099)
Total Hostility	0.337 (0.923)	0.337 (0.922)	0.356 (0.907)	0.324 (0.905)
Woman	1.507* (0.234)	1.500* (0.234)	1.621** (0.217)	1.557** (0.215)
Total Hostility × Woman	0.784 (1.195)	0.798 (1.195)	0.609 (1.174)	0.666 (1.170)
Term-Limited		0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Appointed				0.071** (1.043)
Observations	2909	2980	3038	3062
Clusters	1203	1223	1227	1235
Log Likelihood	-1713.61	-1718.90	-1812.85	-1823.45
AIC	3467.23	3481.79	3669.70	3694.90
McFadden R^2	0.124	0.176	0.175	0.185

Notes: Relative risk ratios are reported with standard errors in parentheses. Exit is the reference outcome. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

4 Separating general and gendered hostility

Table 13: Multinomial Logit Models of Political Ambition

	Model 1	Model 2
Campaign for Reelection vs. Exit		
Visibility	1.074***	1.153***
	(0.024)	(0.038)
Fundraising	1.179**	1.167*
	(0.084)	(0.085)
Democrat	1.278*	1.234
	(0.129)	(0.131)
Legislative Effectiveness	1.114	1.112
	(0.068)	(0.068)
Nonwhite	1.274	1.292
	(0.167)	(0.169)
Year First Elected	1.273***	1.274***
	(0.051)	(0.052)
Female Political Culture	0.952	0.952
	(0.084)	(0.085)
Woman	1.166	1.102
	(0.131)	(0.144)
General Hostility		0.269**
		(0.604)
Gendered Hostility		0.221
		(1.713)
General Hostility × Woman		4.273
		(1.135)
Gendered Hostility × Woman		0.012
		(4.801)
Campaign for Different Office vs. Exit		
Visibility	1.006	1.121**
	(0.035)	(0.055)
Fundraising	0.933	0.921
	(0.120)	(0.121)
Democrat	0.911	0.872
	(0.200)	(0.203)
Legislative Effectiveness	1.086	1.079
	(0.096)	(0.096)
Nonwhite	1.560*	1.556*
	(0.235)	(0.237)
Year First Elected	1.548***	1.538***
	(0.097)	(0.096)
Female Political Culture	1.113	1.119
	(0.120)	(0.121)
Woman	1.512**	1.441*
	(0.176)	(0.198)
General Hostility		0.155*
		(0.970)
Gendered Hostility		0.054
		(3.339)
General Hostility × Woman		0.676
		(1.776)
Gendered Hostility × Woman		20.880
		(6.213)
Observations	2616	2616
Clusters	1059	1059
Log Likelihood	-1807.483	-1801.039
AIC	3650.966	3654.077

Notes: Relative risk ratios are reported with clustered bootstrap standard errors in parentheses (10,000 replications, clustered by individual). Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

5 Abuse 50

Table 14: Total hostility and political ambition: Abuse 50 threshold

	Six Months	Campaign	Lifetime
Likelihood of Campaign for Reelection vs. Exit			
Total Hostility	0.279**	0.110***	0.189**
	(0.549)	(0.657)	(0.717)
Total Hostility × Woman	2.098	8.692	31.057**
	(1.217)	(1.472)	(1.709)
Visibility	1.209***	1.008	0.891**
	(0.048)	(0.048)	(0.049)
Fundraising	1.079	1.114	1.154*
	(0.081)	(0.077)	(0.076)
Democrat	1.060	1.119	1.237
	(0.165)	(0.169)	(0.169)
Legislative Effectiveness	1.052	1.077	1.106
	(0.077)	(0.078)	(0.079)
Nonwhite	1.177	1.298	1.369
	(0.206)	(0.204)	(0.204)
Year First Elected	1.375***	1.382***	1.392***
	(0.062)	(0.062)	(0.062)
Female Political Culture	0.895	0.910	0.930
	(0.104)	(0.105)	(0.106)
Woman	1.138	0.924	0.817
	(0.229)	(0.259)	(0.265)
Likelihood of Campaign for Different Office vs. Exit			
Total Hostility	0.550	0.681	0.992
	(0.858)	(0.942)	(1.012)
Total Hostility × Woman	1.746	7.683	19.457
	(1.582)	(1.807)	(2.037)
Visibility	1.425***	1.133*	1.057
	(0.067)	(0.067)	(0.069)
Fundraising	0.930	0.984	1.011
	(0.103)	(0.100)	(0.098)
Democrat	0.749	0.873	0.929
	(0.237)	(0.241)	(0.241)
Legislative Effectiveness	1.034	1.068	1.086
	(0.113)	(0.113)	(0.113)
Nonwhite	1.252	1.423	1.473
	(0.280)	(0.278)	(0.278)
Year First Elected	1.618***	1.640***	1.660***
	(0.115)	(0.115)	(0.116)
Female Political Culture	1.059	1.091	1.092
	(0.150)	(0.149)	(0.149)
Woman	1.611	1.303	1.198
	(0.322)	(0.350)	(0.353)
Observations	1967	1967	1967
Clusters	938	938	938
Log Likelihood	-1304.004	-1307.817	-1303.017
AIC	2652.007	2659.634	2650.035

Notes: Relative risk ratios are reported with cluster-robust standard errors in parentheses. Standard errors are clustered at the individual level. Exit is the reference outcome. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

6 Abuse 75

Table 15: Total hostility and political ambition: Abuse 75 threshold

	Six Months	Campaign	Lifetime
Likelihood of Campaign for Reelection vs. Exit			
Total Hostility	0.198**	0.115***	0.167**
	(0.689)	(0.815)	(0.868)
Total Hostility × Woman	1.932	9.215	43.998*
	(1.605)	(1.915)	(2.202)
Visibility	1.207***	0.994	0.887**
	(0.048)	(0.048)	(0.049)
Fundraising	1.080	1.121	1.157*
	(0.081)	(0.077)	(0.076)
Democrat	1.066	1.165	1.260
	(0.165)	(0.167)	(0.168)
Legislative Effectiveness	1.055	1.083	1.107
	(0.077)	(0.078)	(0.078)
Nonwhite	1.192	1.302	1.364
	(0.206)	(0.204)	(0.204)
Year First Elected	1.373***	1.381***	1.389***
	(0.062)	(0.062)	(0.062)
Female Political Culture	0.890	0.907	0.928
	(0.104)	(0.105)	(0.106)
Woman	1.178	1.019	0.911
	(0.218)	(0.242)	(0.247)
Likelihood of Campaign for Different Office vs. Exit			
Total Hostility	0.618	0.721	0.917
	(1.062)	(1.163)	(1.222)
Total Hostility × Woman	1.219	9.114	27.734
	(2.046)	(2.327)	(2.602)
Visibility	1.416***	1.132*	1.063
	(0.066)	(0.067)	(0.068)
Fundraising	0.932	0.985	1.008
	(0.103)	(0.100)	(0.098)
Democrat	0.758	0.878	0.922
	(0.236)	(0.239)	(0.239)
Legislative Effectiveness	1.037	1.069	1.083
	(0.113)	(0.113)	(0.113)
Nonwhite	1.262	1.418	1.462
	(0.280)	(0.278)	(0.277)
Year First Elected	1.616***	1.639***	1.656***
	(0.115)	(0.115)	(0.115)
Female Political Culture	1.056	1.087	1.089
	(0.150)	(0.149)	(0.150)
Woman	1.707*	1.407	1.304
	(0.304)	(0.326)	(0.328)
Observations	1967	1967	1967
Clusters	938	938	938
Log Likelihood	-1303.657	-1310.587	-1304.324
AIC	2651.313	2665.173	2652.647

Notes: Relative risk ratios are reported with cluster-robust standard errors in parentheses. Standard errors are clustered at the individual level. Exit is the reference outcome. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

7 Abuse 90

Table 16: Total hostility and political ambition: Abuse 90 threshold

	Six Months	Campaign	Lifetime
Likelihood of Campaign for Reelection vs. Exit			
Total Hostility	0.102**	0.075**	0.114*
	(0.990)	(1.058)	(1.143)
Total Hostility × Woman	2.059	43.092	248.772
	(2.262)	(2.959)	(3.419)
Visibility	1.206***	0.980	0.879***
	(0.047)	(0.047)	(0.048)
Fundraising	1.081	1.130	1.163**
	(0.081)	(0.077)	(0.075)
Democrat	1.072	1.199	1.282
	(0.164)	(0.165)	(0.166)
Legislative Effectiveness	1.055	1.088	1.109
	(0.077)	(0.078)	(0.078)
Nonwhite	1.186	1.304	1.360
	(0.206)	(0.204)	(0.204)
Year First Elected	1.376***	1.382***	1.388***
	(0.062)	(0.062)	(0.062)
Female Political Culture	0.886	0.907	0.928
	(0.105)	(0.105)	(0.106)
Woman	1.205	1.026	0.963
	(0.203)	(0.223)	(0.230)
Likelihood of Campaign for Different Office vs. Exit			
Total Hostility	0.417	0.519	0.643
	(1.542)	(1.536)	(1.641)
Total Hostility × Woman	0.417	33.279	98.538
	(2.992)	(3.552)	(4.016)
Visibility	1.433***	1.137*	1.073
	(0.066)	(0.066)	(0.067)
Fundraising	0.929	0.983	1.003
	(0.103)	(0.100)	(0.098)
Democrat	0.749	0.870	0.904
	(0.236)	(0.236)	(0.236)
Legislative Effectiveness	1.036	1.068	1.080
	(0.113)	(0.113)	(0.113)
Nonwhite	1.256	1.416	1.456
	(0.280)	(0.278)	(0.277)
Year First Elected	1.620***	1.637***	1.651***
	(0.115)	(0.115)	(0.115)
Female Political Culture	1.046	1.083	1.084
	(0.150)	(0.149)	(0.150)
Woman	1.839**	1.440	1.392
	(0.283)	(0.301)	(0.305)
Observations	1967	1967	1967
Clusters	938	938	938
Log Likelihood	-1303.856	-1311.693	-1305.228
AIC	2651.713	2667.386	2654.456

Notes: Relative risk ratios are reported with cluster-robust standard errors in parentheses. Standard errors are clustered at the individual level. Exit is the reference outcome. Inactive cases are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

C Establishing threshold for meaningful effect from the literature

Table 17: Effect Sizes in Related Research on Political Ambition and Exit

Study	Outcome	Effect Size
Håkansson (2024)	Considering leaving politics	+5 pp women vs. men (MPs); +1 pp (municipal)
Håkansson and Josefsson (2026)	Withdrawing from gender-equality debates	+4 pp women vs. men
McCrain and O'Connell (2023)	Running for Congress	+4 pp per additional term; +6 pp for greater legislative professionalization
Ramachandran et al. (2024)	Reduced political ambition following abuse	+13–14 pp women vs. men[a]

Note: Effect sizes are percentage-point (pp) differences or changes in predicted probabilities reported in prior research.

[a] Descriptive gender differences.

D Minimum detectable effect for H1

1 Total Hostility

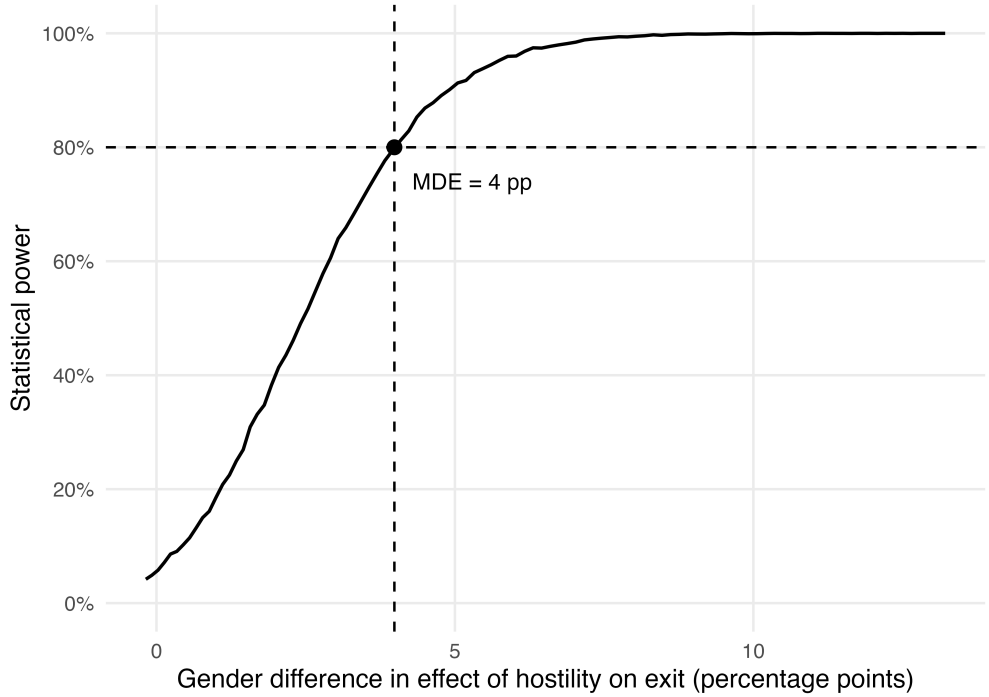
Because a statistically insignificant interaction may reflect limited statistical power, I conduct a simulation-based power analysis to determine the minimum detectable effect (MDE) for **H1**. The MDE identifies the smallest gender difference in the effect of hostility that the analysis can detect with 80% power. Because **H1** predicts that hostility disproportionately increases women's likelihood of voluntary exit, I use a one-sided test with $\alpha = .05$.

The power analysis follows the specification of the multinomial logit model used to test **H1**. The outcome distinguishes between exiting politics, campaigning for reelection, and campaigning for a different office, with exit as the reference category. I retain the observed sample and covariate values and vary the size of the interaction between Total Hostility and Woman. For each imposed interaction, I calculate the resulting difference between women and men in the effect of hostility on the predicted probability of exit. Effects are evaluated over an interquartile-range increase in Total Hostility, equivalent to a 14.3 percentage-point increase in the proportion of hostile tweets received.

To account for repeated observations of legislators, the analysis uses the sampling distribution from 10,000 bootstrap replications clustered at the legislator level. For each imposed effect size, I use this distribution to estimate the probability that the analysis would detect an effect in the direction predicted by **H1**. I then increase the imposed effect until statistical power reaches 80%. Because the multinomial model estimates separate contrasts for campaigning for reelection versus exit and campaigning for a different office versus exit, the simulation imposes the same interaction on both contrasts and uses their joint bootstrap distribution when evaluating statistical power.

Figure 10 presents the resulting power curve. The analysis reaches 80% power when an interquartile-range increase in hostility has an effect on women's probability of exit that is approximately 4 percentage points larger than its effect on men's probability of exit. Thus, the estimated MDE is approximately 4 percentage points. Effects smaller than this threshold cannot be reliably detected with the present data, but the analysis is well powered to detect gender differences of approximately 4 percentage points or larger.

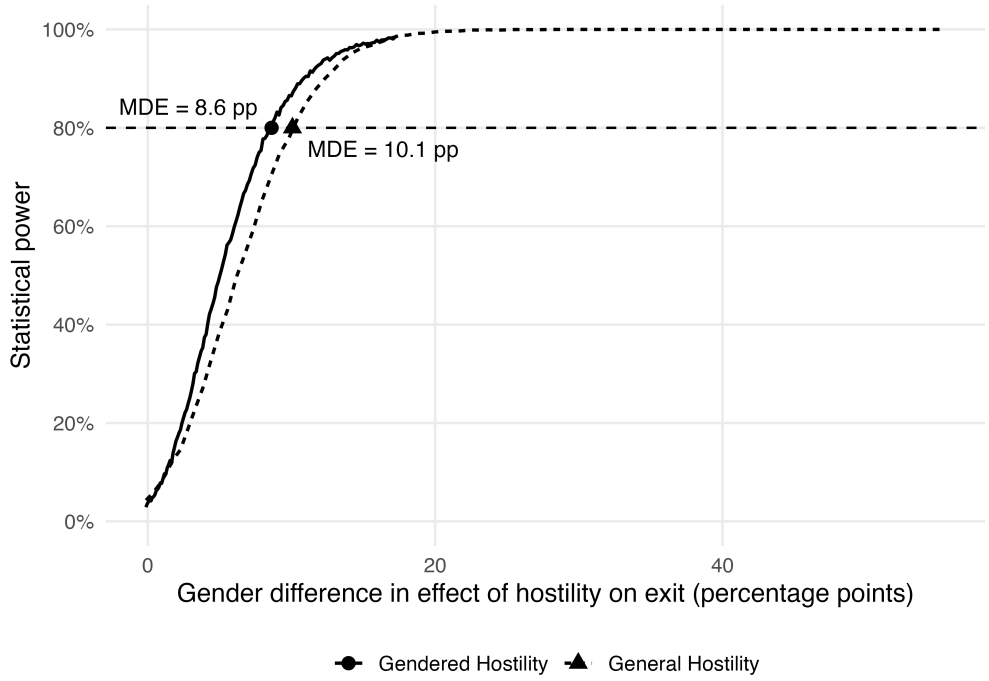
Figure 10: H1 analysis is powered to detect minimum effect of 4 percentage points



Note: The figure shows statistical power to detect the gender difference in the effect of hostility on voluntary exit. Effect sizes on the x-axis represent the difference between women and men in the change in predicted probability of exit associated with an interquartile-range increase in Total Hostility (14.3 percentage points). Power is estimated using the sampling distribution from 10,000 bootstrap replications clustered at the legislator level and a one-sided test with $\alpha = .05$. The dashed lines identify 80% power and the corresponding minimum detectable effect of approximately 4 percentage points.

2 General and Gendered Hostility

Figure 11: H1 analysis is less powered to detect small effects for gendered and general hostility separately



Note: The figure shows statistical power to detect gender differences in the effects of general and gendered hostility on voluntary exit. Effect sizes on the x-axis represent the difference between women and men in the change in predicted probability of exit associated with an interquartile-range increase in each type of hostility. Interquartile ranges are calculated among legislators exposed to each type of hostility and correspond to increases of 19.9 percentage points for general hostility and 5.0 percentage points for gendered hostility. Power is estimated using the sampling distribution from 10,000 bootstrap replications clustered at the legislator level and a one-sided test with $\alpha = .05$. The horizontal dashed line identifies 80% power. The corresponding minimum detectable effects are approximately 10.1 percentage points for general hostility and 8.6 percentage points for gendered hostility.

E Do more ambitious politicians receive more hostility?

1 Twitter behavior doesn't vary between those who exit vs. stay

I match legislators in my dataset with that created by Butler et al. (2023). Butler et al. (2023) used human coders to label tweets by state legislators along several criteria: political ideology,[31] positive or negative sentiment,[32] policy content,[33] statement of opinion,[34] and requests for donation.[35] Tweet count (logged) refers to how often the politician tweeted.

The descriptive scores of behavior on Twitter are gathered prior to the filing deadlines included in the binomial and multinomial analyses. By matching legislators in my dataset with Butler et al. (2023) I am able to compare previous Twitter behavior of the politicians who voluntarily exit within my observed time frame and those who don't. I collapse the dataset to one row per legislator. Table 18 presents descriptive comparisons between legislators who eventually voluntarily exit vs. stay in politics. Binary variables—Democrat, woman, nonwhite—are reported as counts (percentages), while continuous variables are reported as means (standard deviations).

Table 18: Descriptive Statistics by Legislators' Exit Status

Characteristic	Stayed (N = 775)	Exited (N = 432)
Tweet Count (logged)	5.93 (1.85)	6.07 (1.95)
Ideology	0.13 (0.11)	0.13 (0.10)
Sentiment	0.48 (0.24)	0.46 (0.26)
Opinion	0.68 (0.12)	0.70 (0.13)
Asks for Donation	0.00 (0.01)	0.00 (0.00)
No Policy Content	0.66 (0.14)	0.67 (0.14)
Legislative Effectiveness	1.06 (0.89)	1.02 (0.97)
Years Since First Elected	0.10 (0.92)	-0.19 (1.15)
Female Political Culture	0.68 (0.74)	0.75 (0.73)
Democrat	392 (50.6%)	190 (44.0%)
Woman	242 (31.2%)	112 (25.9%)
Nonwhite	183 (23.6%)	65 (15.0%)

Next, I estimate a binomial logistic regression. The dependent variable is whether the politician voluntarily exited politics. The six Twitter behavior measures are entered simultaneously, allowing each characteristic to be evaluated conditional on the others. The model

[31] This score ranges from negative one to one, with larger values representing a higher rate of conservative versus liberal tweeting.

[32] This score ranges from negative one to one, with larger values indicating a higher frequency of positive tweets relative to negative tweets.

[33] Based on the policy categories used by Wilkerson and Adler (2014) to code federal bills.

[34] Proportion of tweets including opinion.

[35] Proportion of tweets requesting donations.

additionally accounts for party, gender, race, legislative effectiveness, years since first election, and the state's political culture toward women. The analysis includes 635 legislators with complete data on the Twitter measures and controls.

Table 19 provides little evidence that legislators who subsequently exit are distinguishable by their prior Twitter behavior. None of the six Twitter characteristics is associated with subsequent exit at conventional levels of statistical significance. I additionally compare the full model to a restricted model containing only the legislator and institutional controls. A likelihood-ratio test indicates that adding the six Twitter behavior measures does not significantly improve model fit ($\chi^2(6) = 8.34$, $p = .214$). Thus, the Twitter characteristics are neither individually nor jointly associated with subsequent political exit.

Table 19: Pre-Filing Twitter Behavior and Subsequent Political Exit

	Exit
Tweet Count (logged)	0.074
	(0.049)
Ideology	−0.721
	(0.931)
Sentiment	−0.320
	(0.399)
Opinion	0.438
	(0.768)
Asks for Donation	−15.113
	(21.555)
No Policy Content	1.306*
	(0.752)
Democrat	−0.208
	(0.214)
Woman	−0.124
	(0.194)
Nonwhite	−0.219
	(0.253)
Legislative Effectiveness	−0.107
	(0.105)
Years Since First Elected	−0.397***
	(0.094)
Female Political Culture	0.143
	(0.126)
AIC	819.849
Log Likelihood	−396.924
Num. obs.	635

Note: Logistic regression coefficients with standard errors in parentheses. The dependent variable equals one if the legislator subsequently exits politics and zero if the legislator remains in office. Twitter behavior is measured prior to the filing deadline. A likelihood-ratio test of the Twitter behavior measures yields $\chi^2(6) = 8.34$, $p = 0.214$. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

2 Targeting politicians based on observable quality

Table 20: Weak correlation between observable quality indicators and total hostility

	Total Hostility
Visibility	0.049***
	(0.005)
Woman	0.001
	(0.055)
Democrat	−0.070***
	(0.013)
Legislative Effectiveness	−0.011*
	(0.005)
Nonwhite	0.004
	(0.014)
Years Since First Elected	−0.006
	(0.007)
Fundraising	−0.020*
	(0.011)
Female Political Culture	−0.010
	(0.008)
Visibility × Woman	−0.003
	(0.008)
Clusters	731
R^2	0.195
Adj. R^2	0.187
Num. obs.	923

*Note:*The dependent variable is total hostility received during the six months prior to the filing deadline. Visibility is the logged number of Twitter mentions received during the same period. OLS coefficients are reported with cluster-robust standard errors clustered by legislator in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

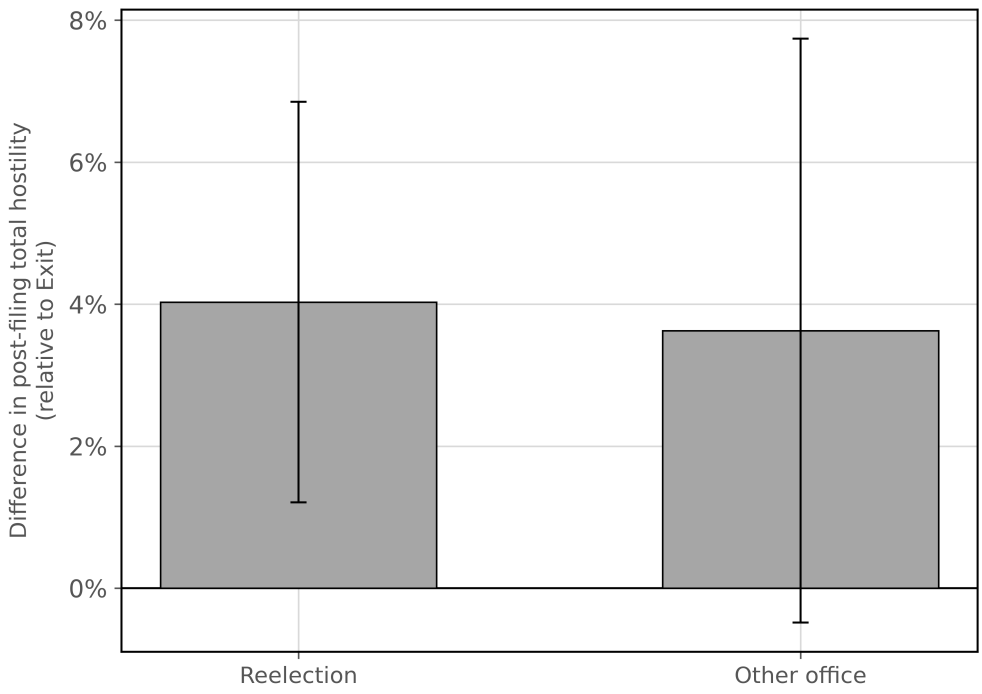
3 Hostility six months prior vs. six months after filing deadline

Table 21: Post-filing total hostility following filing deadlines

	Total hostility
Reelection	0.040**
	(0.014)
Other office	0.036+
	(0.021)
Prior hostility	0.426***
	(0.044)
Visibility	0.014**
	(0.004)
Fundraising	-0.004
	(0.007)
Woman	-0.012
	(0.010)
Democrat	-0.059***
	(0.012)
Nonwhite	0.003
	(0.012)
Observations	815
R^2	0.330

Notes: OLS estimates with standard errors clustered by legislator in parentheses. The dependent variable is the proportion of mentions classified as hostile in the six months after the filing deadline. Total hostility is defined as the proportion of general hostility plus the proportion of gendered hostility. Exit is the reference category for the campaign-decision coefficients. Models include election-year fixed effects.
+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 12: Slight increase in hostility for politicians who campaign for reelection, relative to exit



Note: The y-axis indicates the adjusted difference in the proportion of hostility in mentions by filing decision. Bars represent estimated differences relative to legislators who voluntarily exited. Error bars denote 95% confidence intervals. Predicted values come from Table 21.

F Selection Effects

1 Multinomial, interacting female political culture with total hostility

Table 22: Female political culture moderates relationship between online hostility and exit

	Women
Likelihood of Campaign for Reelection vs. Exit	
Visibility	1.118
	(0.080)
Fundraising	1.062
	(0.188)
Democrat	2.100***
	(0.249)
Legislative Effectiveness	1.082
	(0.139)
Nonwhite	1.225
	(0.305)
Year First Elected	1.323**
	(0.113)
Total Hostility	6.353
	(1.895)
Female Political Culture	0.849
	(0.193)
Total Hostility × Female Political Culture	0.088*
	(1.287)
Likelihood of Campaign for Different Office vs. Exit	
Visibility	1.126
	(0.099)
Fundraising	1.123
	(0.282)
Democrat	1.126
	(0.335)
Legislative Effectiveness	1.097
	(0.154)
Nonwhite	1.461
	(0.394)
Year First Elected	1.433***
	(0.139)
Total Hostility	0.495
	(2.283)
Female Political Culture	1.089
	(0.240)
Total Hostility × Female Political Culture	0.331
	(1.270)
Observations	774
Clusters	308
Log Likelihood	-523.709
AIC	1087.418
McFadden R^2	0.034

Notes: Relative risk ratios are reported with clustered bootstrap standard errors in parentheses. Standard errors are based on 10000 bootstrap replications clustered at the individual level. Exit is the reference outcome. Inactive cases, term-limited legislators, men,and appointed legislators are excluded. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

G Descriptives

1 Race, gender, party breakdown

Table 23: Composition of Legislators in the Sample

	Men		**Women**	
Race	Democrat	Republican	Democrat	Republican
White	225	505	135	126
Nonwhite	135	12	95	11